Statistical Learning: A Mathematical Perspective and Statistical Distributions

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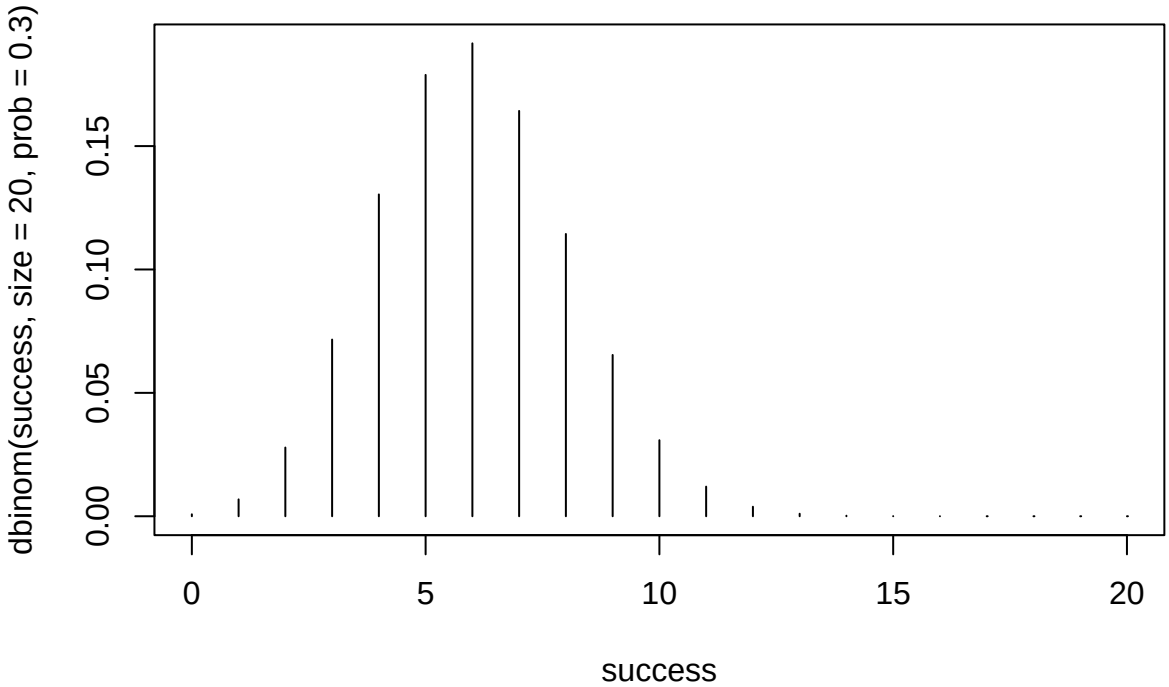
The main goal of this session is to familiarize Biostatistics students with the visualization of different Probability Distributions as would have been covered on first day of class. The distributions will include binomial, Poisson and Normal distributions. Bivariate normal, Poisson and Binomial will also be covered as well as the Dirichlet distributions. The Trivariate normal will be covered assuming some given covariance structure. The second objective is to introduce R/RStudio and RMarkdown to students.

Distributions

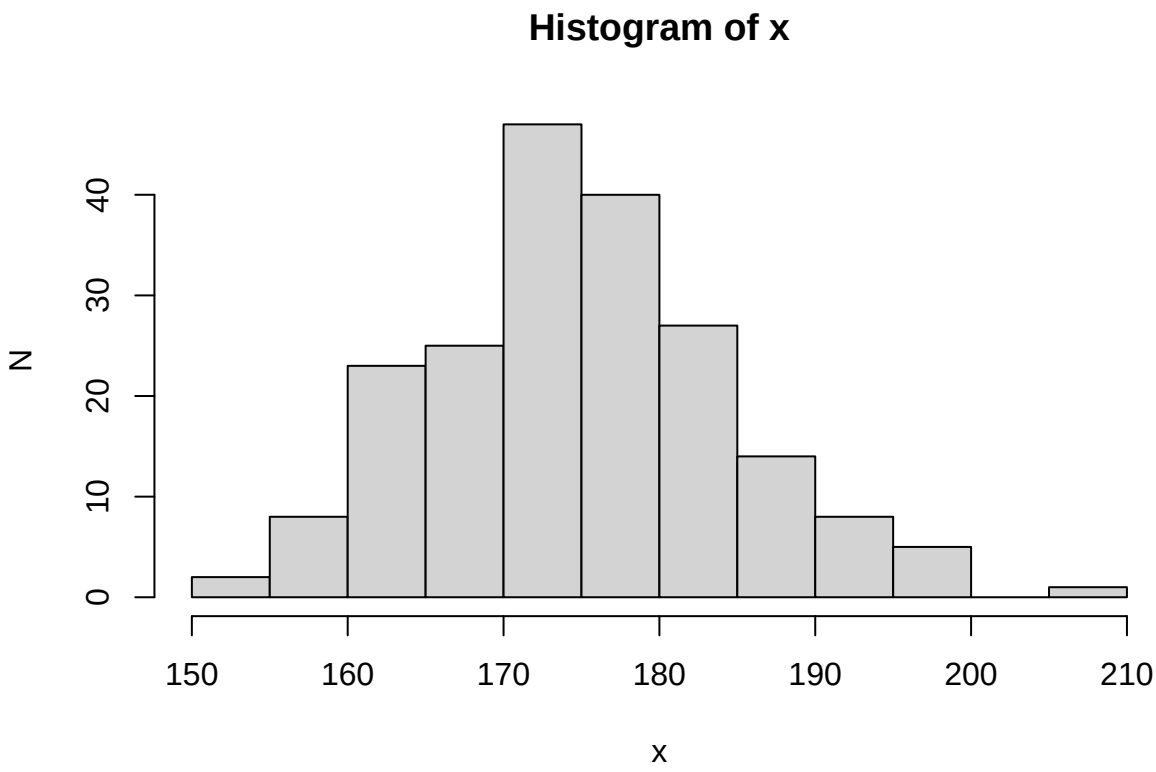
Binomial distribution

```
success <- 0:20

plot(success, dbinom(success, size=20, prob=.3),type='h')
```



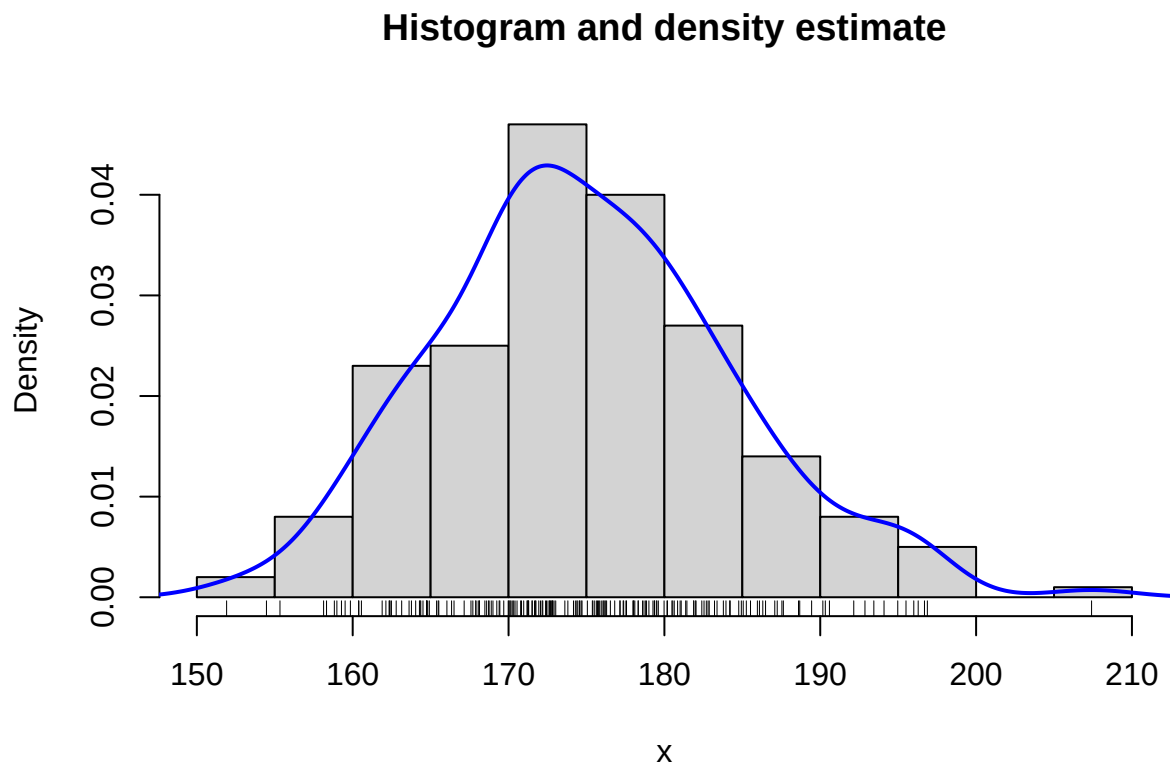
Normal distribution

```
set.seed(123)
x <- rnorm(200, 175, 10)
hist(x, xlab="x", ylab="N", breaks="FD")
```



Add normal probability plot

```
hist(x, freq=FALSE, xlab="x", breaks="FD",
     main="Histogram and density estimate")
lines(density(x), lwd=2, col="blue")
rug(jitter(x))
```

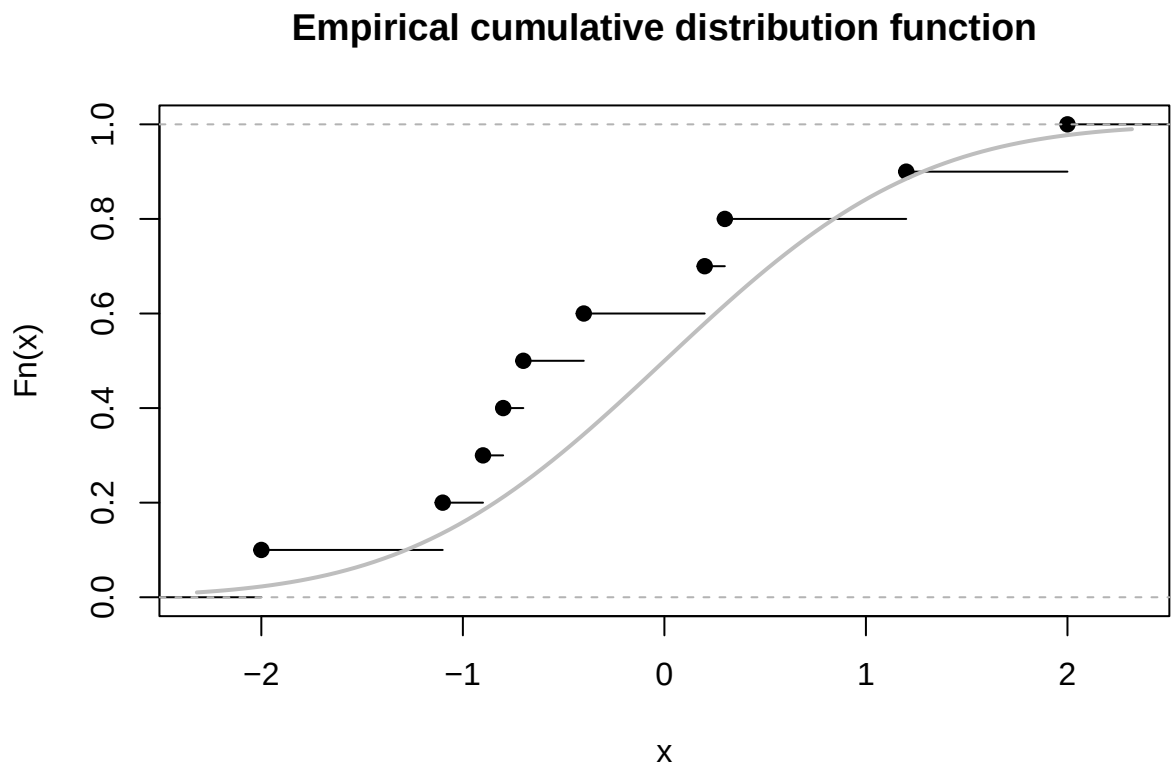


Empirical cumulative distribution (CDf)

```
vec <- round(rnorm(10), 1)
mean(vec)
```

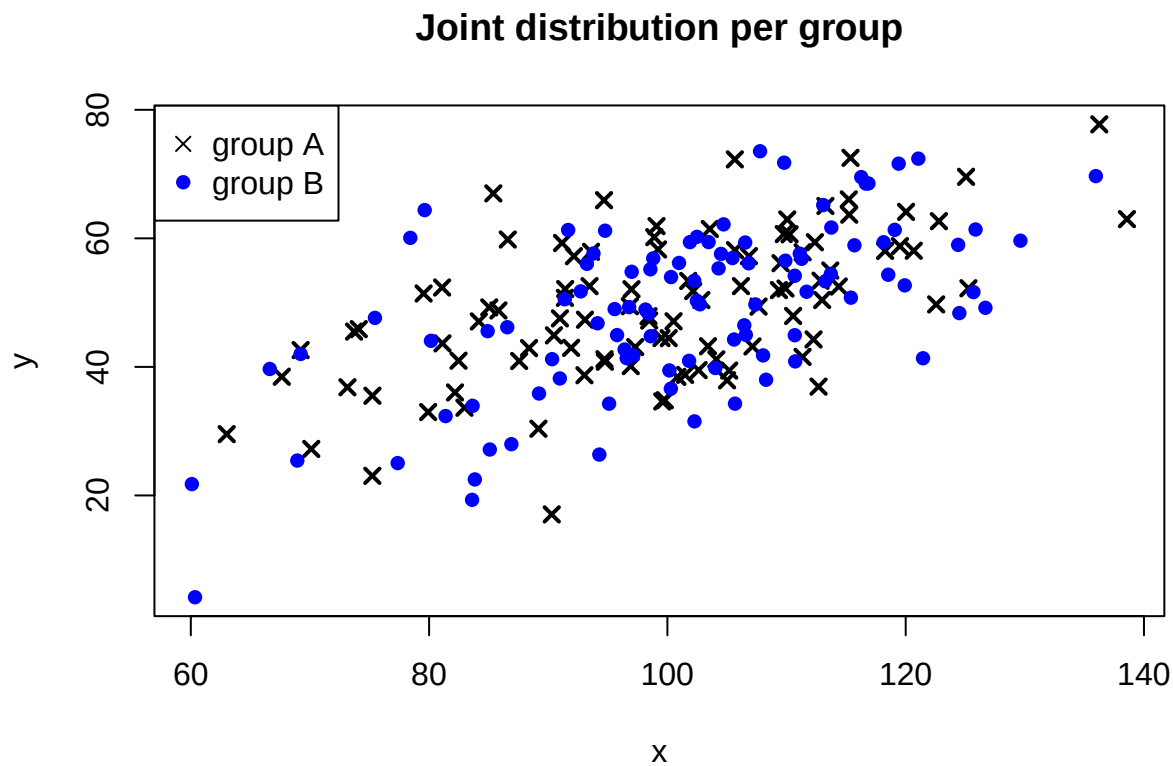
```
## [1] -0.22
```

```
Fn  <- ecdf(vec)
plot(Fn, main="Empirical cumulative distribution function")
curve(pnorm, add=TRUE, col="gray", lwd=2)
```



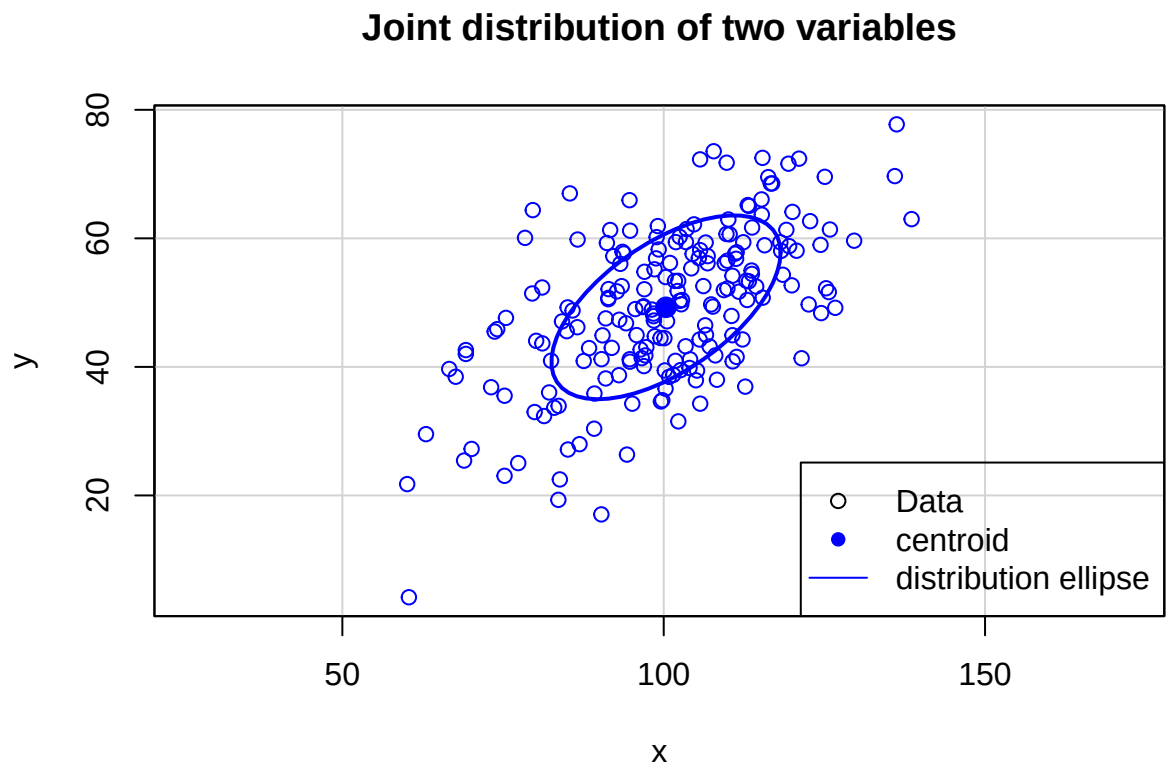
Joint distribution with group membership

```
N  <- 200
P  <- 2
x  <- rnorm(N, 100, 15)
y  <- 0.5*x + rnorm(N, 0, 10)
IV <- gl(P, N/P, labels=LETTERS[1:P])


plot(x, y, pch=c(4, 16)[unclass(IV)], lwd=2,
     col=c("black", "blue")[unclass(IV)],
     main="Joint distribution per group")
legend(x="topleft", legend=c("group A", "group B"),
       pch=c(4, 16), col=c("black", "blue"))
```

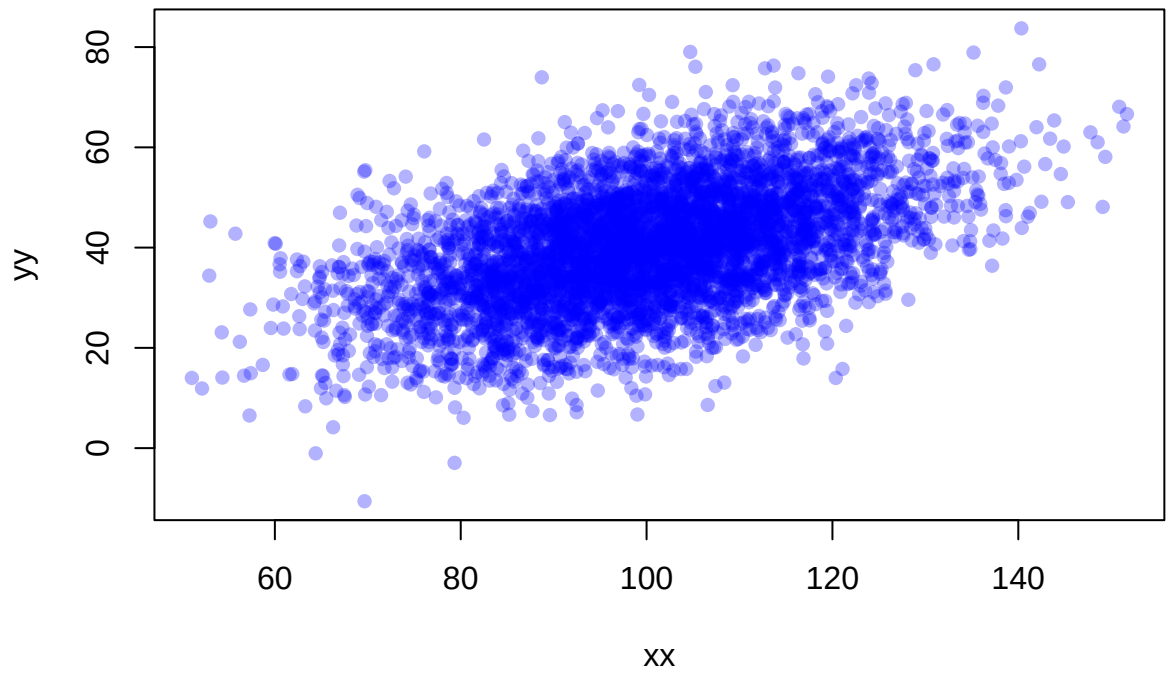


Joint density

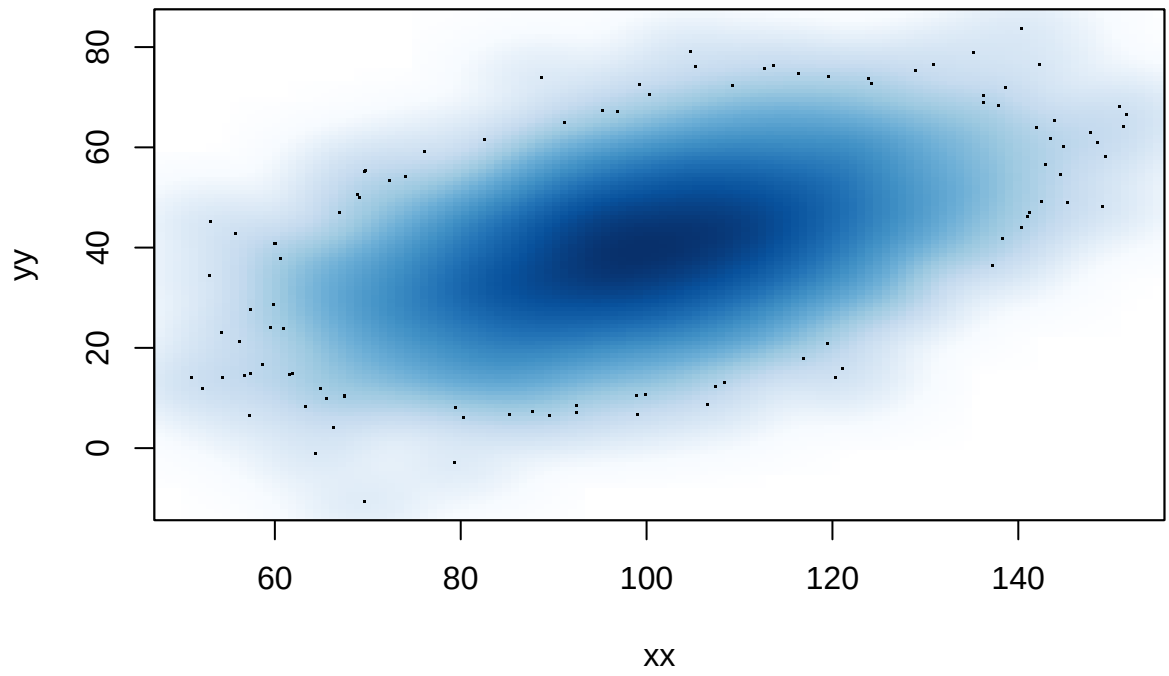
```
library(car)
dataEllipse(x, y, xlab="x", ylab="y", asp=1, levels=0.5, lwd=2, center.pch=16,
            col="blue", main="Joint distribution of two variables")
legend(x="bottomright", legend=c("Data", "centroid", "distribution ellipse"),
       pch=c(1, 16, NA), lty=c(NA, NA, 1), col=c("black", "blue", "blue"))
```



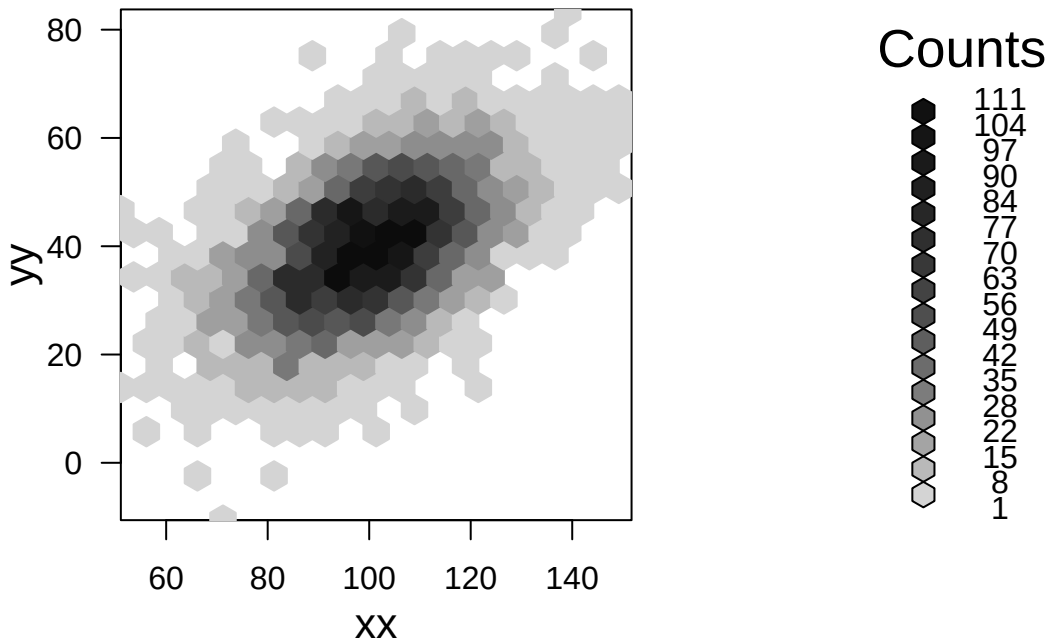

```r
# With many observations

N  <- 5000
xx <- rnorm(N, 100, 15)
yy <- 0.4*xx + rnorm(N, 0, 10)
plot(xx, yy, pch=16, col=rgb(0, 0, 1, 0.3))
```




```r
# Smooth scatter plot
smoothScatter(xx, yy, bandwidth=4)
```




```r
# 2-D plot
library(hexbin)
res <- hexbin(xx, yy, xbins=20)
plot(res)
```



#Bivariate distributions

Discrete Bivariate Uniform Distribution

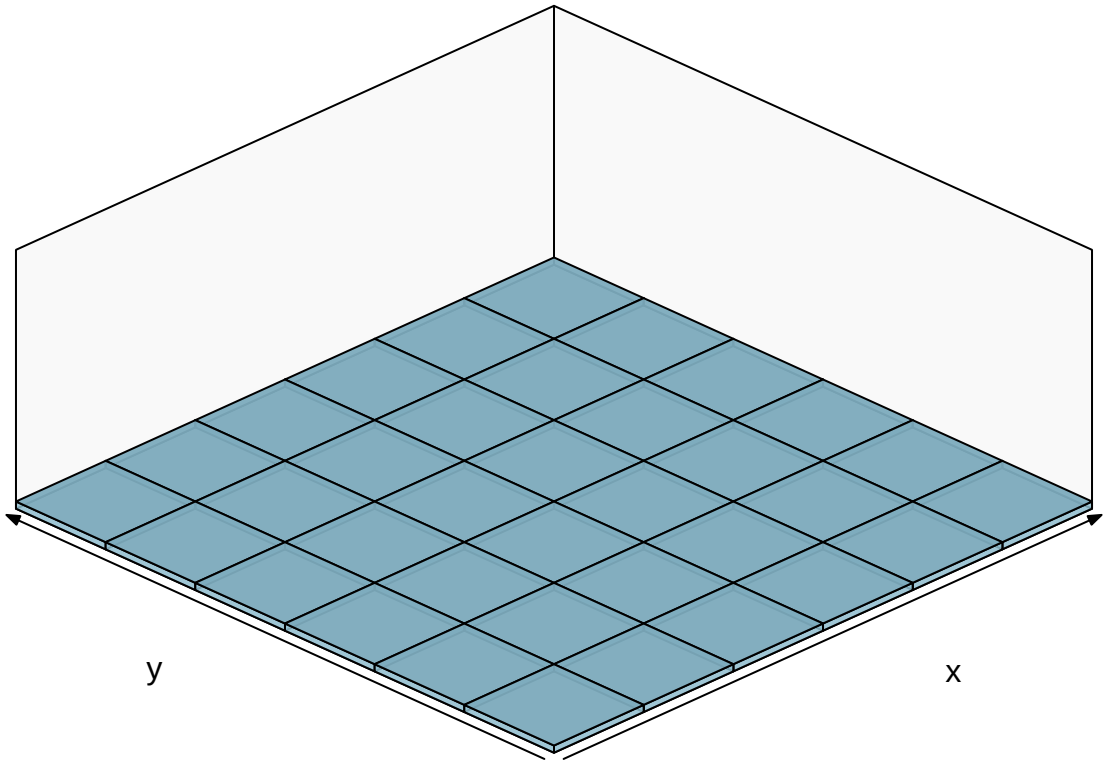
We can describe a discrete bivariate uniform distribution as the product of two discrete univariate uniform distributions, so:

$$P(X = x, Y = y) = f_{XY}(x, y; a_X, b_X, a_Y, b_Y) = f_X(x; a_X, b_X) f_Y(a_Y, b_Y)$$

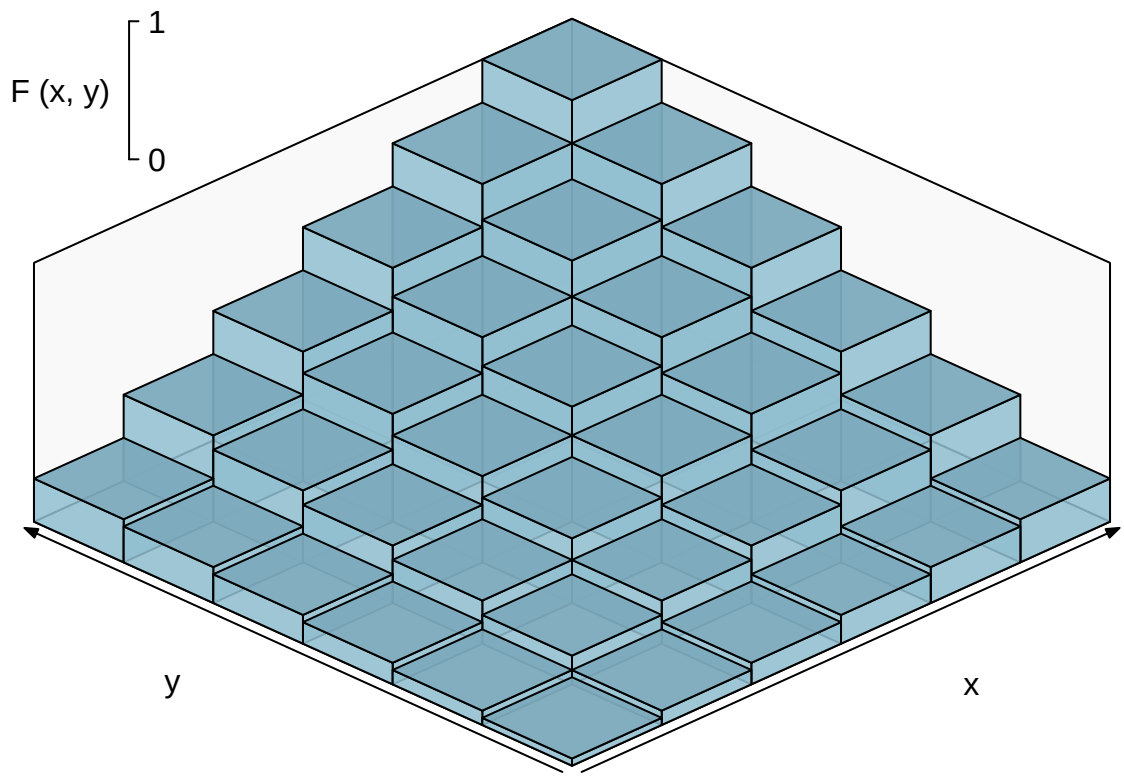
$$= \left[\frac{1}{b_X - a_X + 1}\right]\left[\frac{1}{b_Y - a_Y + 1}\right]$$

where a_X and b_X are the integers giving the minimum and maximum possible values for X. Likewise a_Y and b_Y are minimum and maximum for Y. Example: Let $(x, y) \in [1, 6]$, then the joint density is visualized as below:

```
f <- dubvpmf (
1, 6, #first variable
1, 6) #second variable
G <- dubvcdf (
1, 6,
1, 6)

plot(f)
```




```
#Cumulative joint density function

plot(G)
```




```
#Evaluating joint density and cumulative probabilities

f(6,6)
```

```
## [1] 0.02777778
```

```
G(6,6)
```

```
## [1] 1
```

Bivariate Binomial Distributions

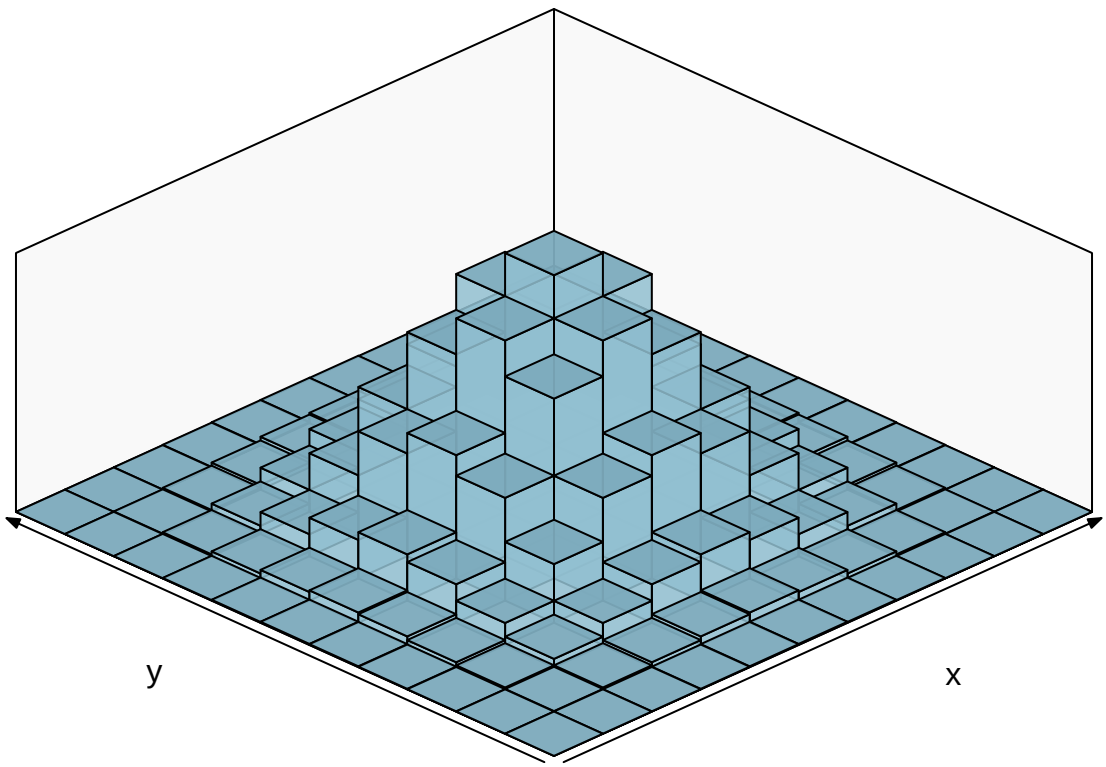
One way to deefine a bivariate binomial distribution is to say that we have n trials. In each trial there are two independent events, each with a particular probability of success. Like flipping two coins, n times. The joint density is given as:

$$P(X = x, Y = y) = f_{XY}(x, y; p_X, p_Y, n) = f_X(x; p_X, n) f_Y(y; p_Y, n)$$

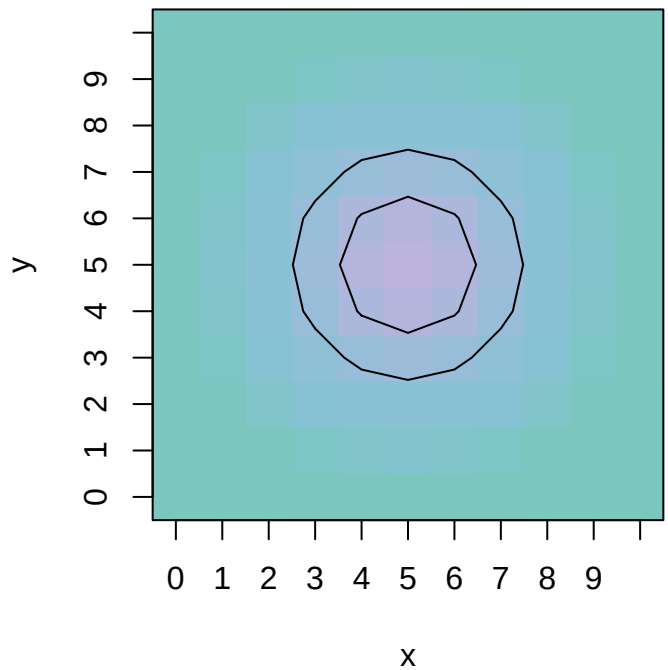
$$\left[\binom{n}{x} p_X^x (1 - p_X)^{n-x}\right] \left[\binom{n}{y} p_Y^y (1 - p_Y)^{n-y}\right]$$

where $p_X,\ and\ p_Y$ are probabilities of success X and Y respectively. Example and visualization

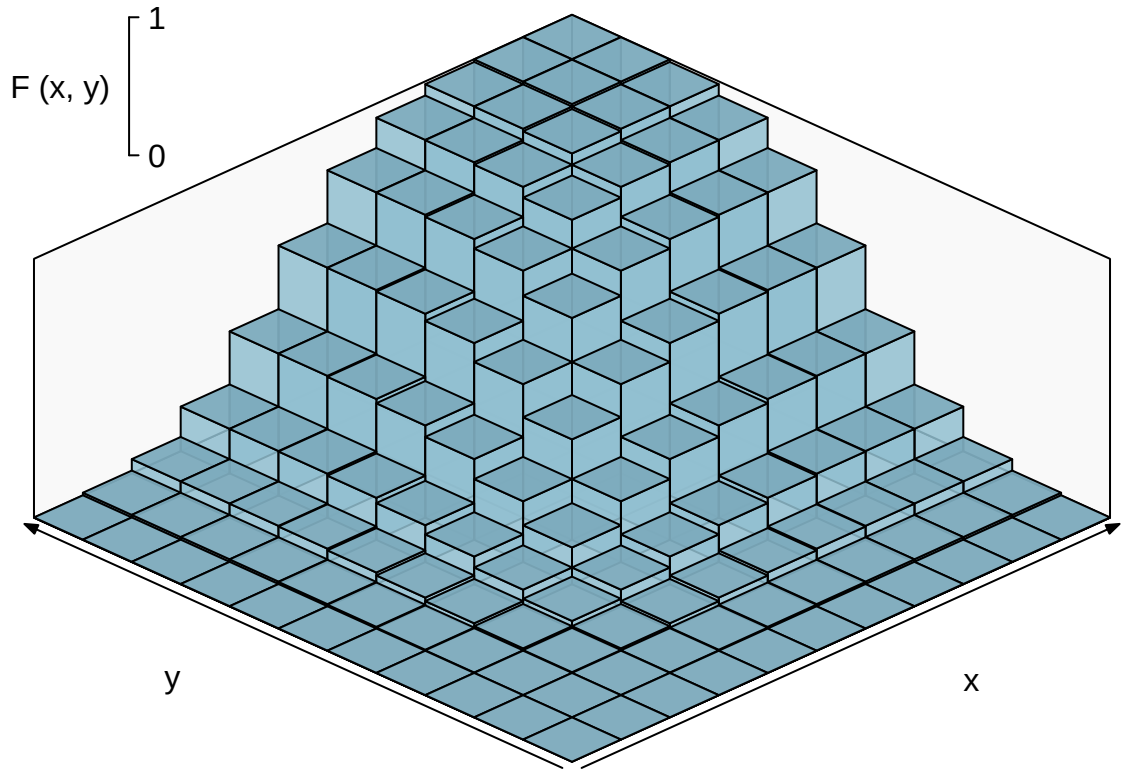
```r
f <- bnbvpmf (0.5, 0.5, 10)
G <- bnbvcdf (0.5, 0.5, 10)

#visualize

plot(f)
```




```r
#visualizing the contours for the bivariate distribution 2-D
plot(f, FALSE)
```




```
#visualizing the cumulative joint binomial density function
plot(G)
```



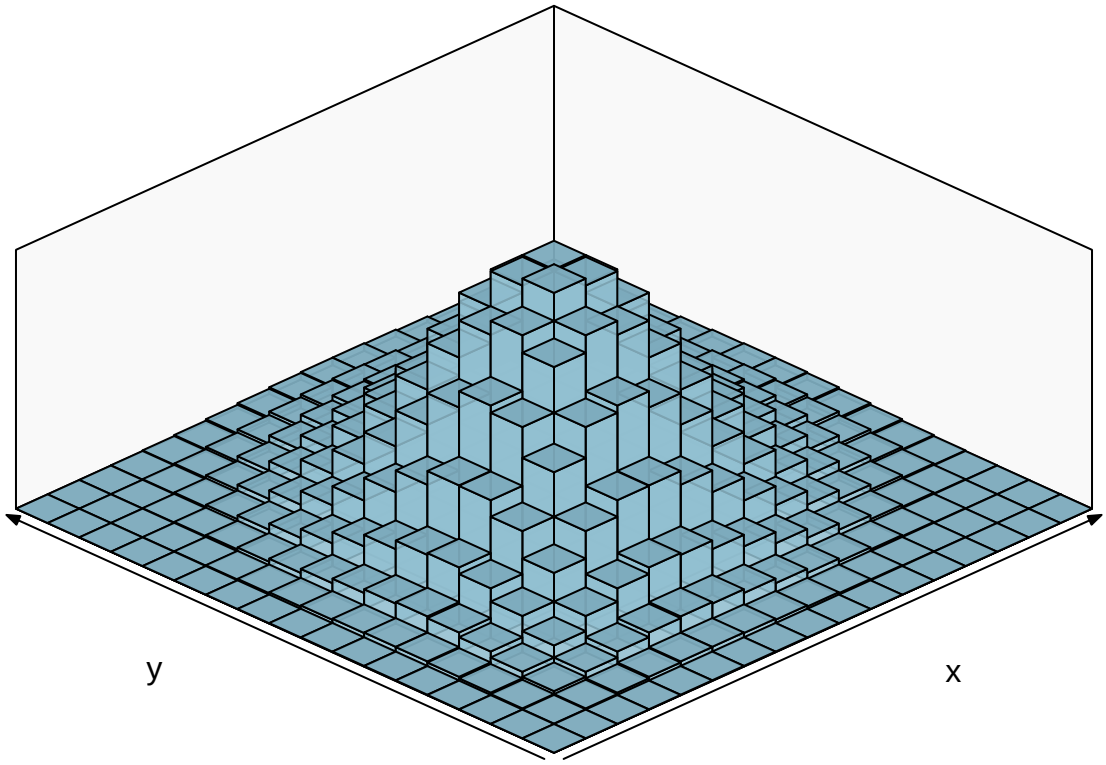
Bivariate Poisson Distributions

According to Karlis and Ntzoufras (2003), we can define the bivariate Poisson probability mass function as:

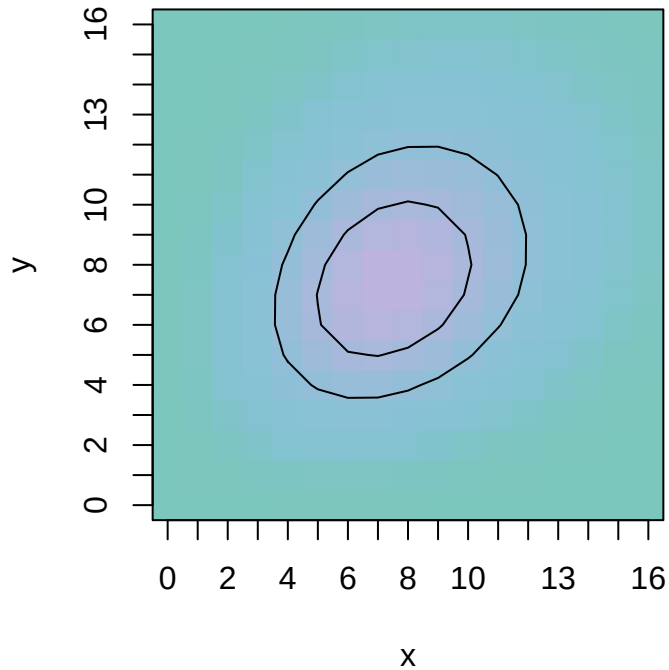
$$P(X = x, Y = y) = f_{XY}(x, y; \lambda_1, \lambda_2, \lambda_3)$$
$$= e^{-(\lambda_1+\lambda_2+\lambda_3)} \frac{\lambda_1^x}{x!} \frac{\lambda_2^y}{y!} \sum_k \binom{x}{k} \binom{y}{k} k! \left(\frac{\lambda_3}{\lambda_1 \lambda_2}\right)^k$$

where $\lambda_1, \lambda_2, \lambda_3$ are positive real integers and k is the sequence in 0 to min(X,Y) and $E(X) = Var(X) = \lambda_1 + \lambda_3$, $E(Y) = Var(Y) = \lambda_2 + \lambda_3$ and $Cov(X, Y) = \lambda_3$. As you may have noted, this joint probability distribution is not a product of the marginals. Example: consider joint Poisson distribution where $E(X) = E(Y) = 8$ and $Cov(X, Y) = 2$, then:

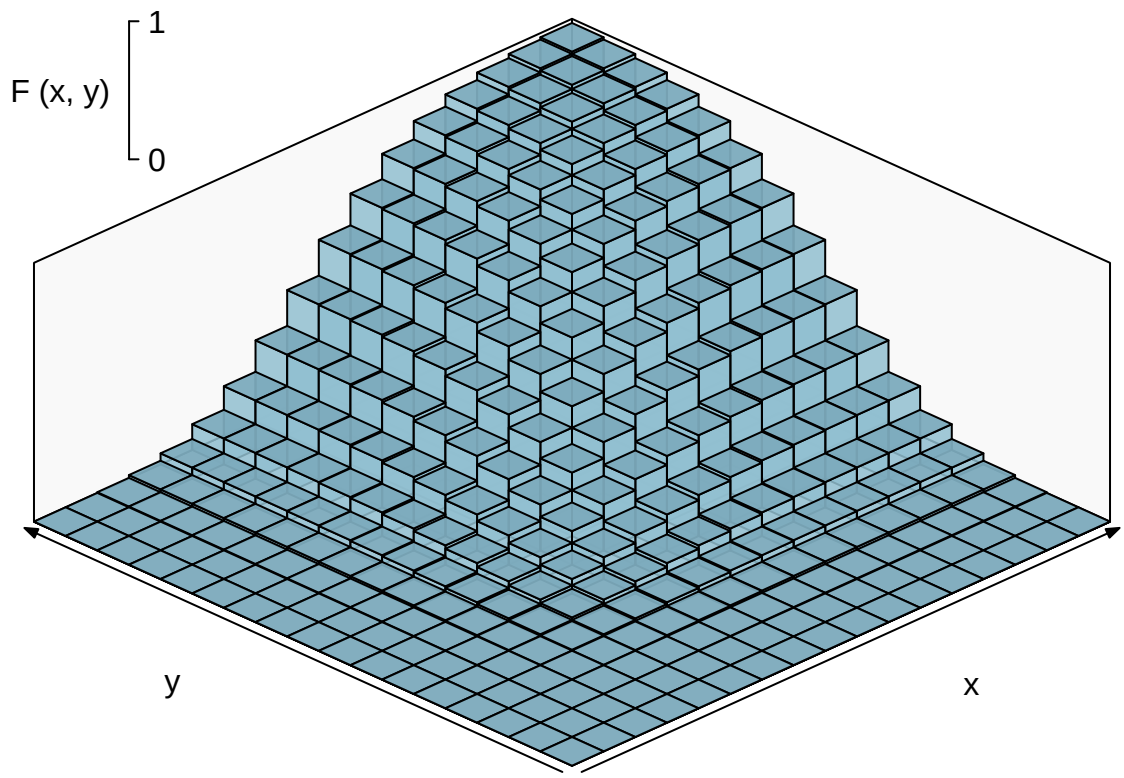
```
f <- pbvpmf.2 (8, 8, 2)

#3-D
plot (f)
```




```
#2-D
plot (f, FALSE)
```




```
#CDF
G<-pbvcdf.2(8,8,2)
plot(G)
```



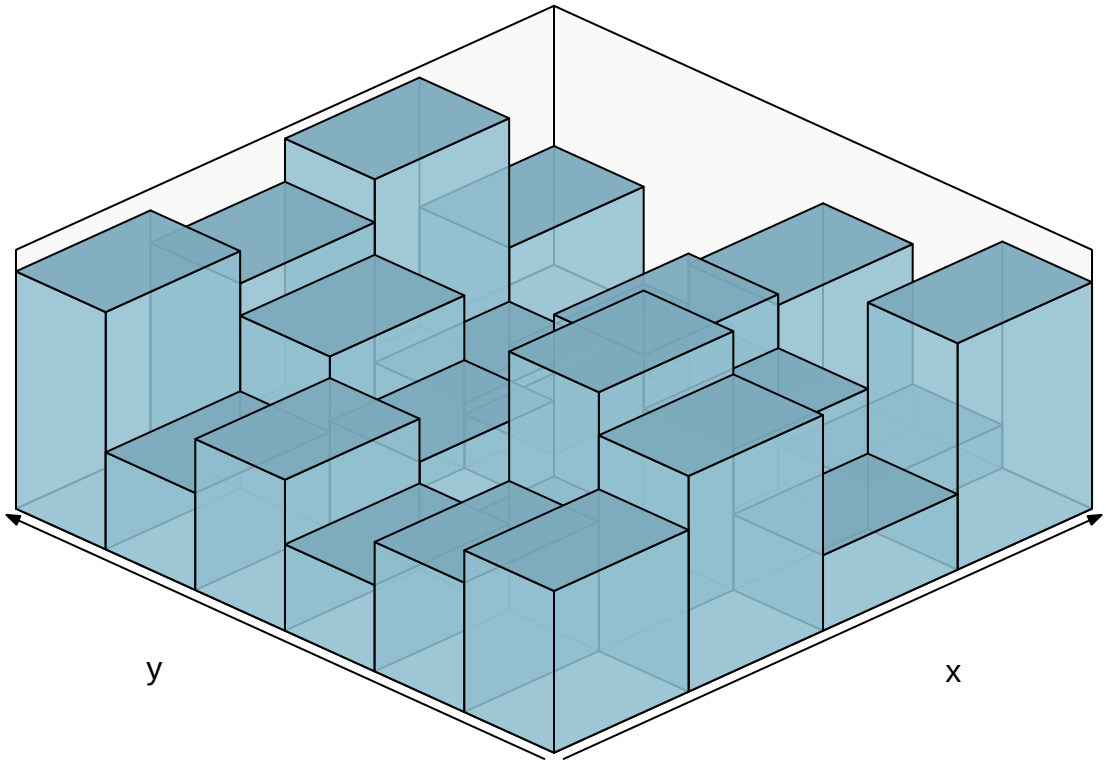
Bivariate Categorical Distributions

Construct probability mass functions for categorical distribution eg multinomial etc. It takes a single argument, a matrix of probabilities (or frequencies), preferably with row and column names. Here's an example using a matrix of random counts:

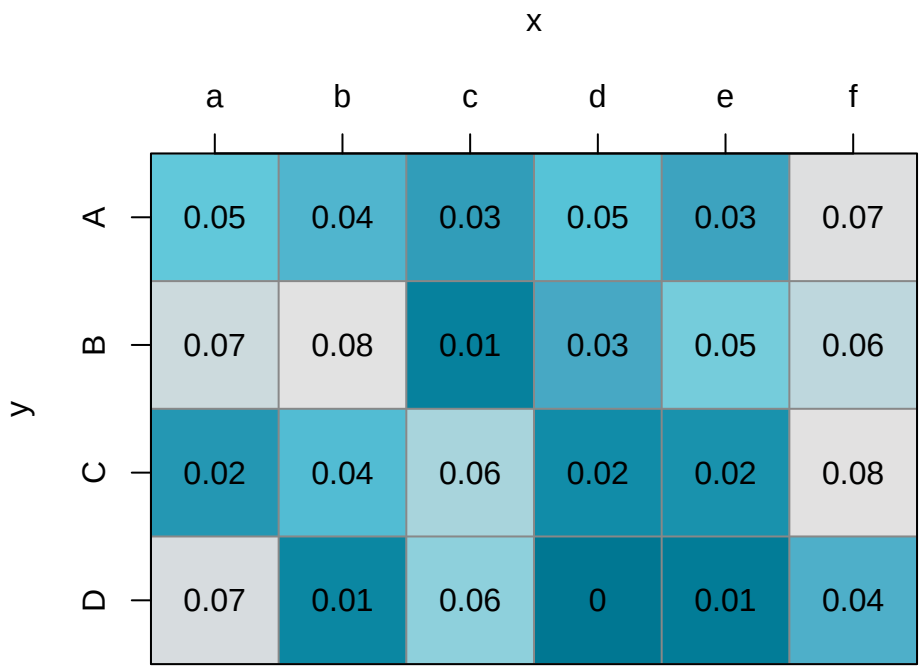
```
h <- matrix (sample (1:24), 4, 6)

rownames (h) <- LETTERS [1:4]
colnames (h) <- letters [1:6]
h
```

```
##    a  b  c  d  e  f
## A 15 12  8 14  9 22
## B 20 24  3 10 16 19
## C  7 13 18  5  6 23
## D 21  4 17  1  2 11
```

```
f <- gbvpmf (h)

#3-D visualization
plot(f)
```




```r
#2-D visualization
plot(f, FALSE)
```




```
#Note that the 2d plot has a different orientation from other 2d plots

#Joint probablity can be obtained by evaluating:
h [2, 4] / sum (h)
```

```
## [1] 0.03333333
```

```
#or
f(2,4)
```

```
## [1] 0.03333333
```

```
#or
f ("B", "d")
```

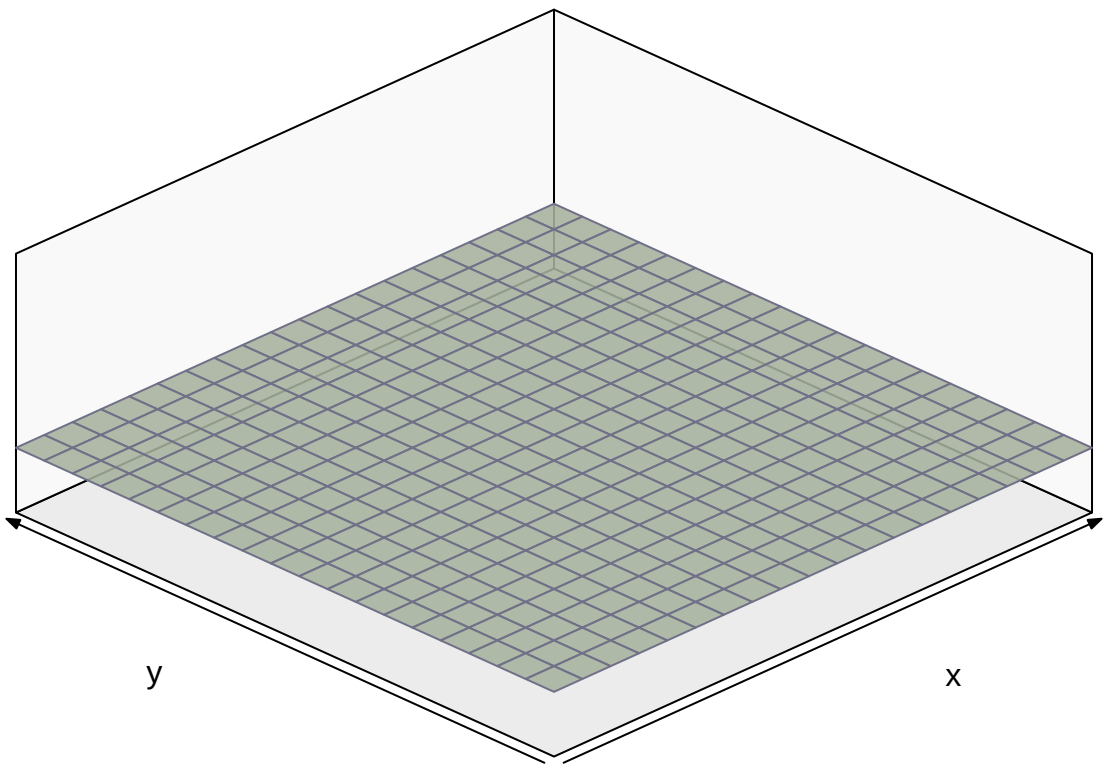
```
## [1] 0.03333333
```

Continuous Bivariate Uniform Distributions

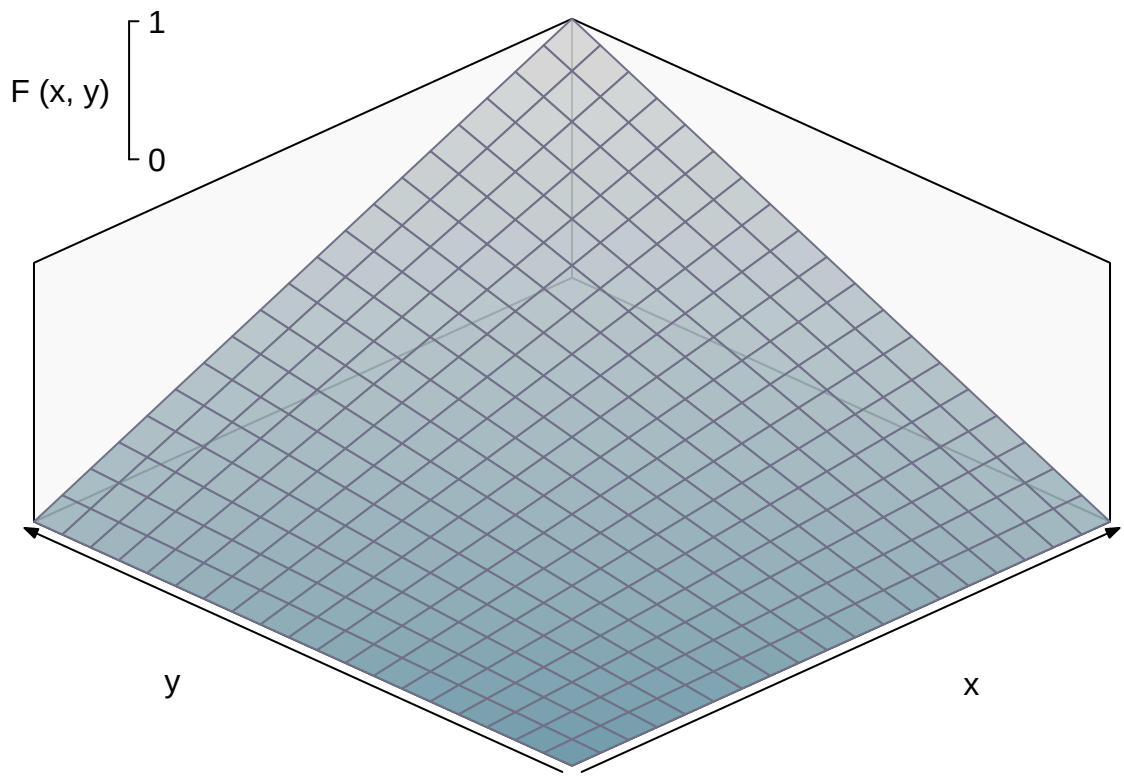
Continuous bivariate uniform distributions are similar to discrete bivariate uniform distri- butions. However, we have a probability density function rather than a probability mass function.

Here's an example, where both X and Y , can take values between zero and two:

```r
f <- cubvpdf (
0, 2, #first variable
0, 2) #second variable
G <- cubvcdf (
0, 2,
0, 2)

#And the plots of the functions
plot(f)
```




```r
plot(G)
```

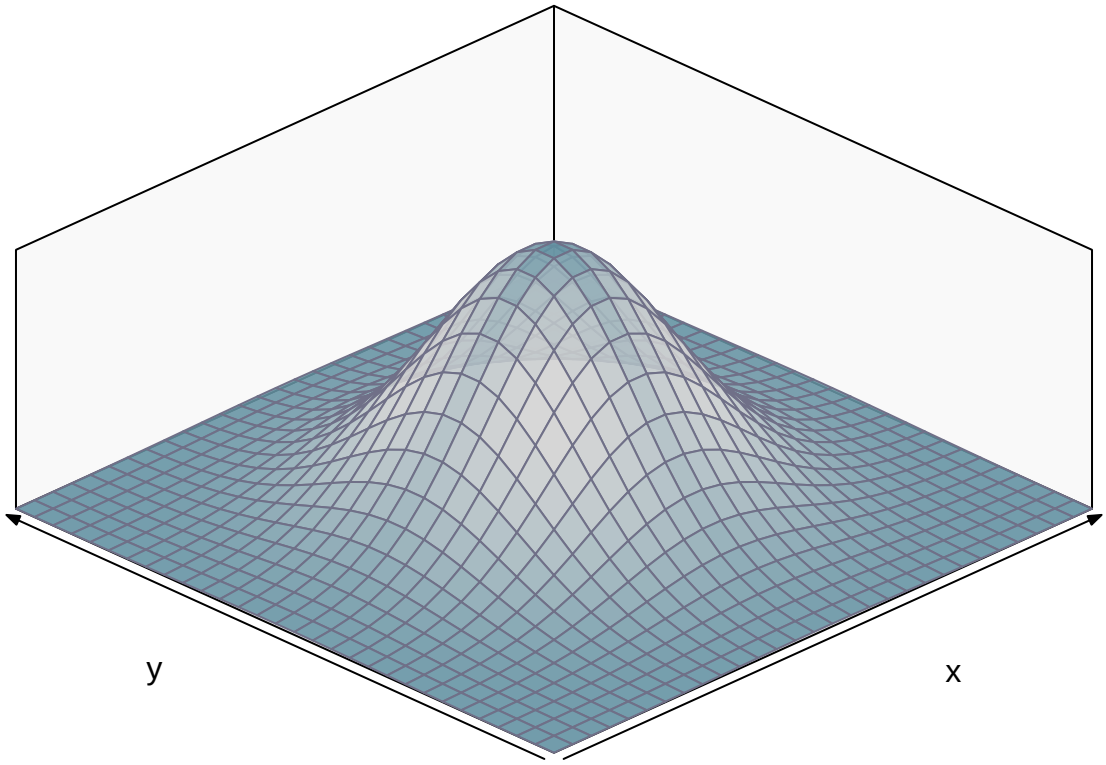


Bivariate Normal Distributions

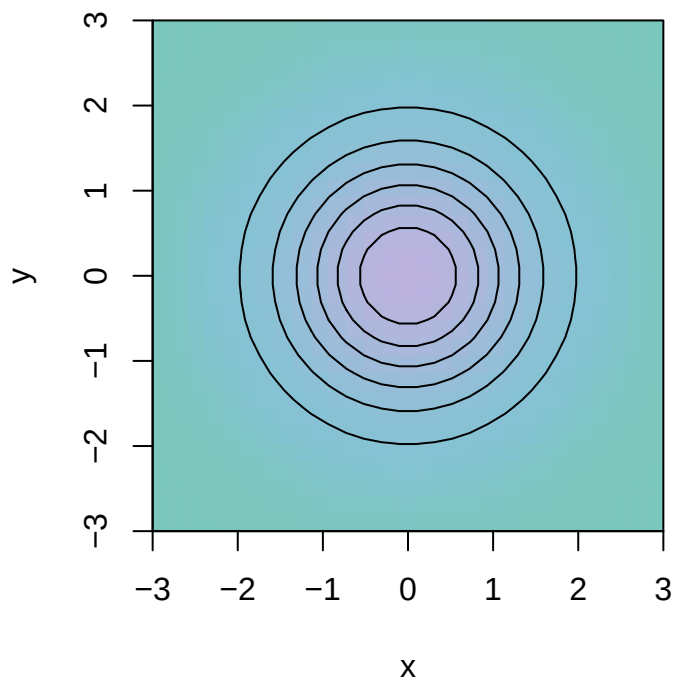
All functions take five parameters. The firrst functions (with no suffix) take the means of X and Y , the standard deviations of X and Y , and their correlation. The second functions (with the suffix) take the means of X and Y , the variances of X and Y , and their covariance.

Here's an example with zero means, standard deviations of one, and no correlation: (Essentially, a bivariate generalization of the standard normal distribution").

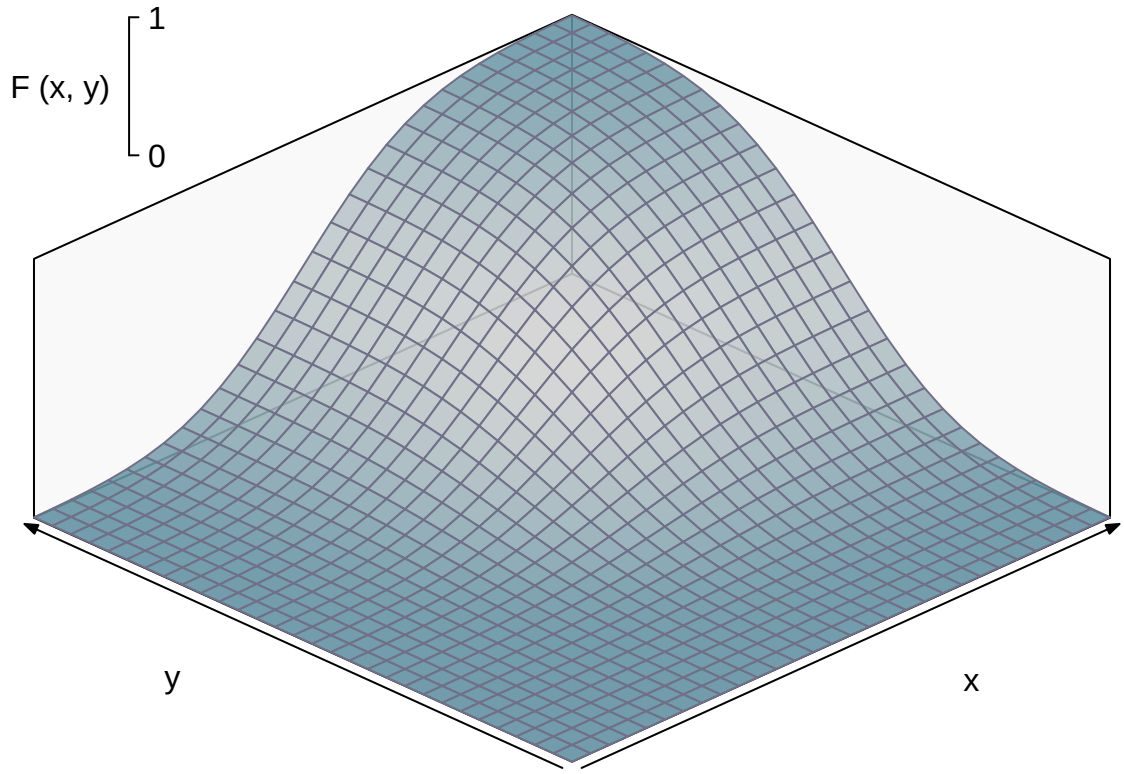
```
f <- nbvpdf (0, 0, 1, 1, 0)
G <- nbvcdf (0, 0, 1, 1, 0)

#3-D
plot (f)
```



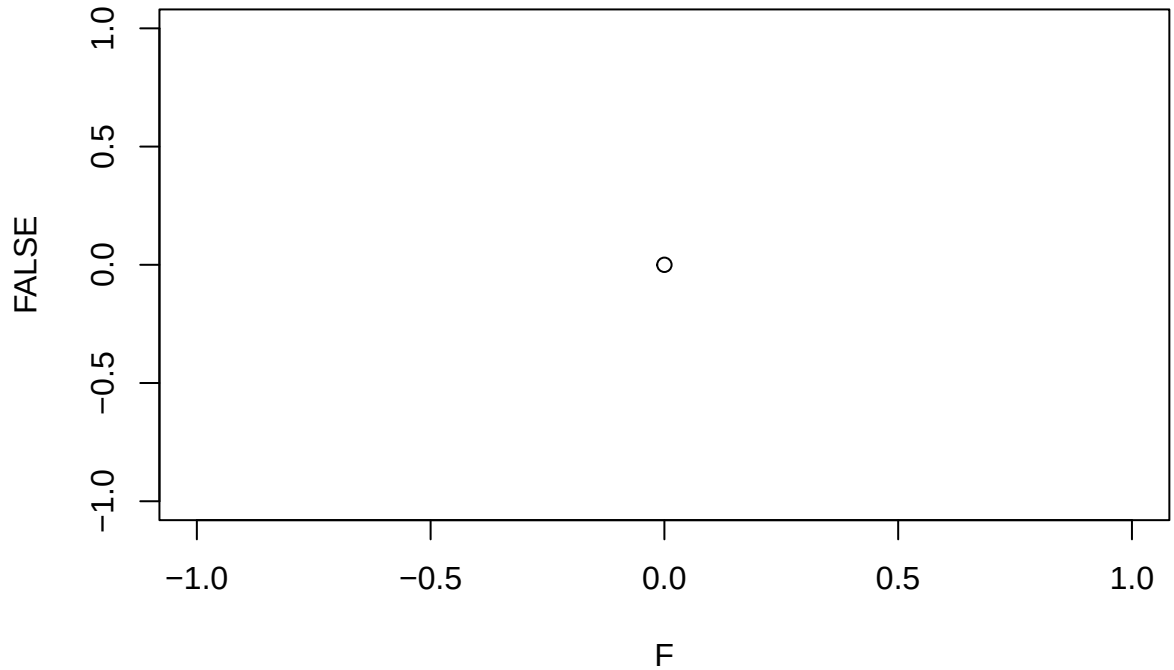

```
#2-D
plot (f, FALSE)
```




```
#CDF
plot (G)
```




```
plot (F, FALSE)
```

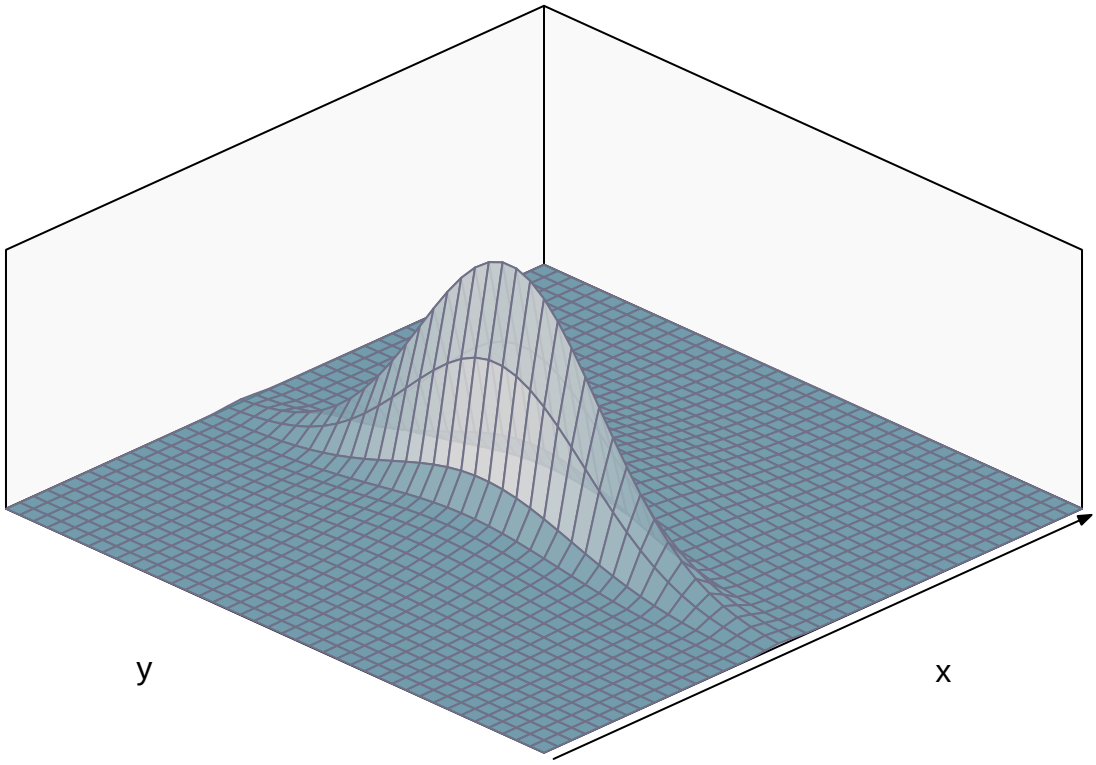


Bivariate Bimodal Distributions

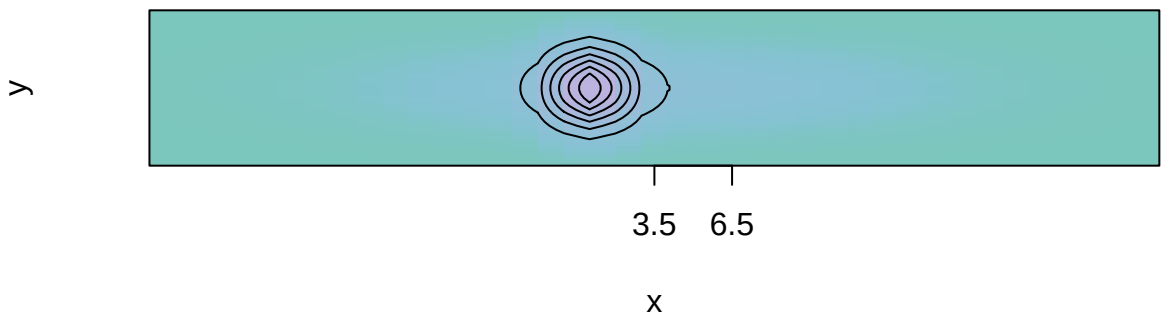
It's possible to construct a bivariate bimodal probability density function by taking two bivariate normal probability density functions, then adding their densities together, and then dividing by two.

The first four arguments are the means and standard deviations (or variances) of the first component distribution. The last four arguments are the means and standard deviations (or variances) of the second component distribution.Here's a an example, where the component means of X are 3.5 and 6.5, the component means of Y are zero, and all standard deviations are one:

```
f <- bmbvpdf (
3.5, 1, 1, 1, #first component distribution
6.5, 1, 1, 1) #second component distribution

#3-D
plot (f,
axes = c (TRUE, FALSE), xat = c (3.5, 6.5) )
```



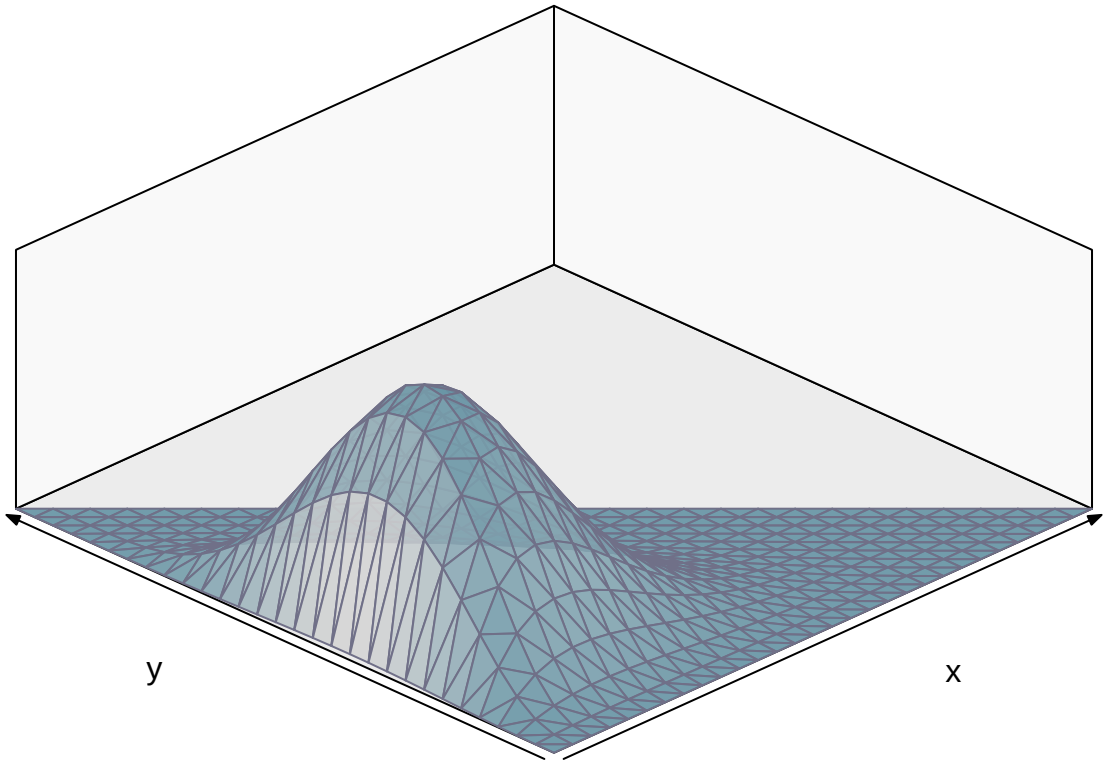

```
#2-D
plot (f, FALSE,
axes = c (TRUE, FALSE), xat = c (3.5, 6.5) )
```



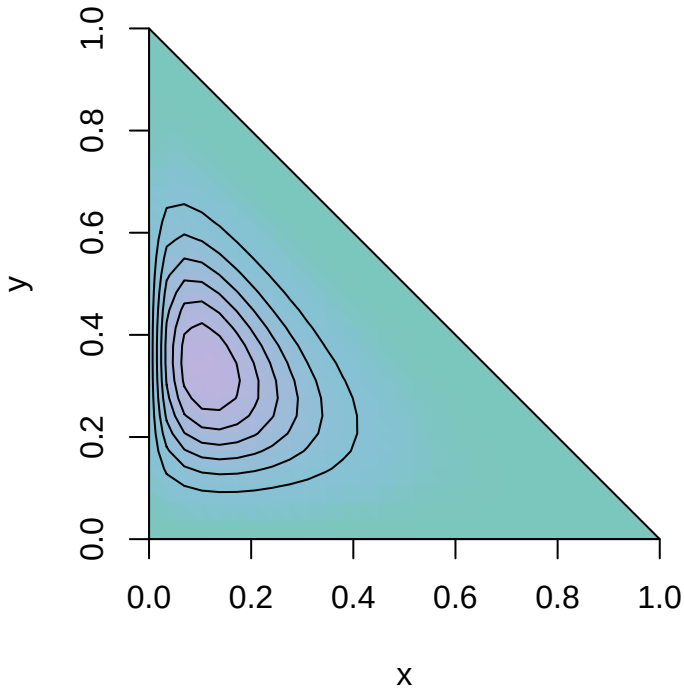
Trivariate Dirichlet Distributions

Dirichlet distributions with three variables are similar to other probability distributions with two variables.(Because it's possible to compute the third variable from the first two variables). The probability density function takes three α parameters. Here's an example with α parameters of two, four and six:

```
f <- dtvpdf (2, 4, 6)

plot(f)
```



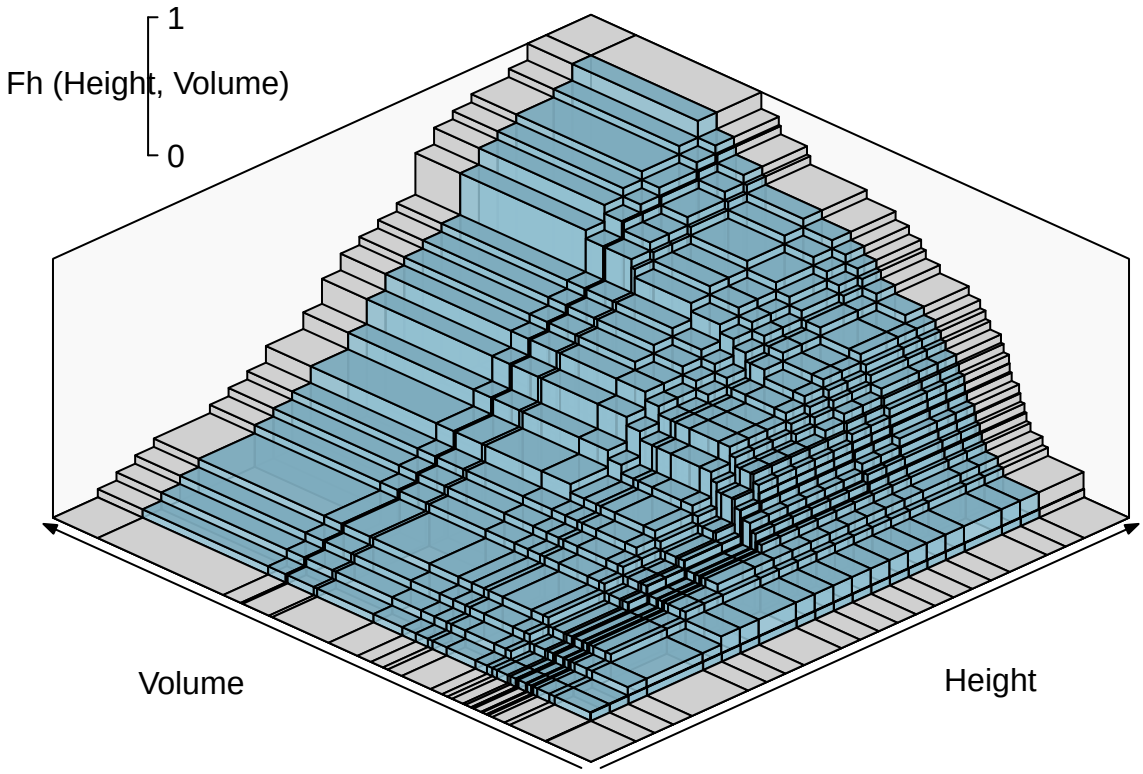

```
#2-D

plot(f, FALSE)
```



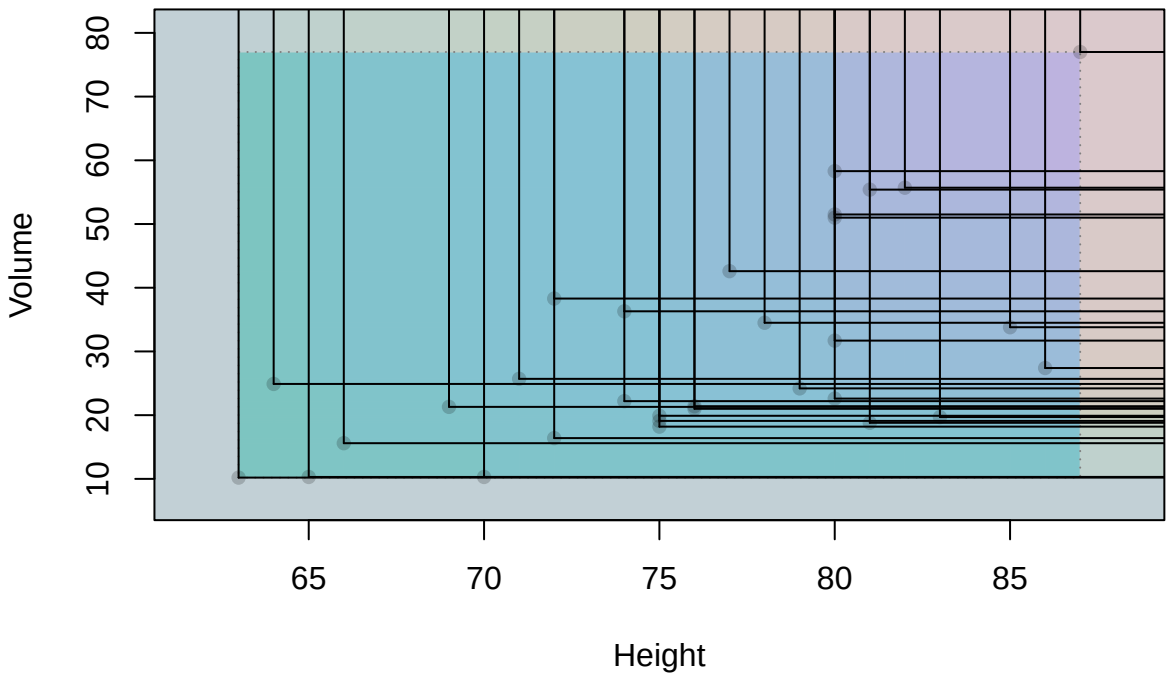
Bivariate Empirical Cumulative Distribution Functions

Like bivariate kernel density estimates, bivariate ECDFs are computed from vectors of data. The resulting probability distributions are step functions (so, have discrete properties), however in general, they represent the distributions of continuous random variables. Here's an example, using the trees data:

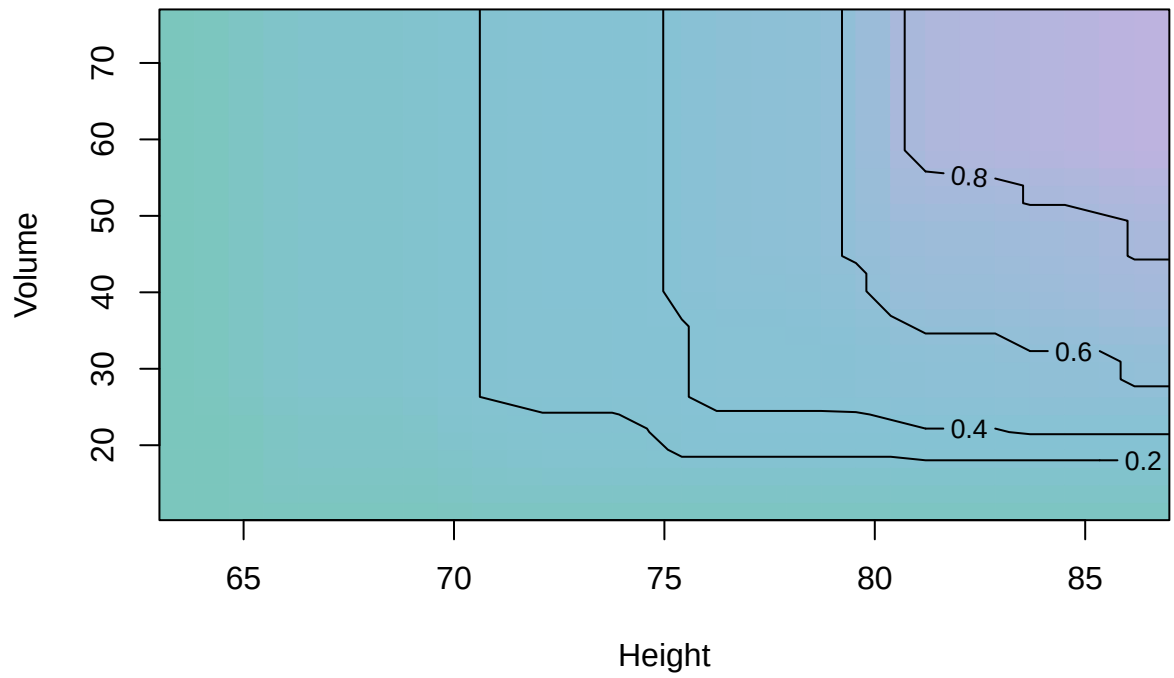
```
attach (trees)
Fh <- ebvcdf (Height, Volume)
detach (trees)


plot (Fh)
```




```
#2-D
plot (Fh, FALSE)
```




```
plot (Fh, FALSE, reg=TRUE)
```



Note that by default, if the number of observations is less than or equal to forty, the function is plotted as a (discrete) step function, with an extrapolated region outside the range of observed values. However, if the number of observations is larger than forty, the function is plotted as a (continuous) surface, evaluated over a regularly-spaced grid.

Trivariate Normal Distributions

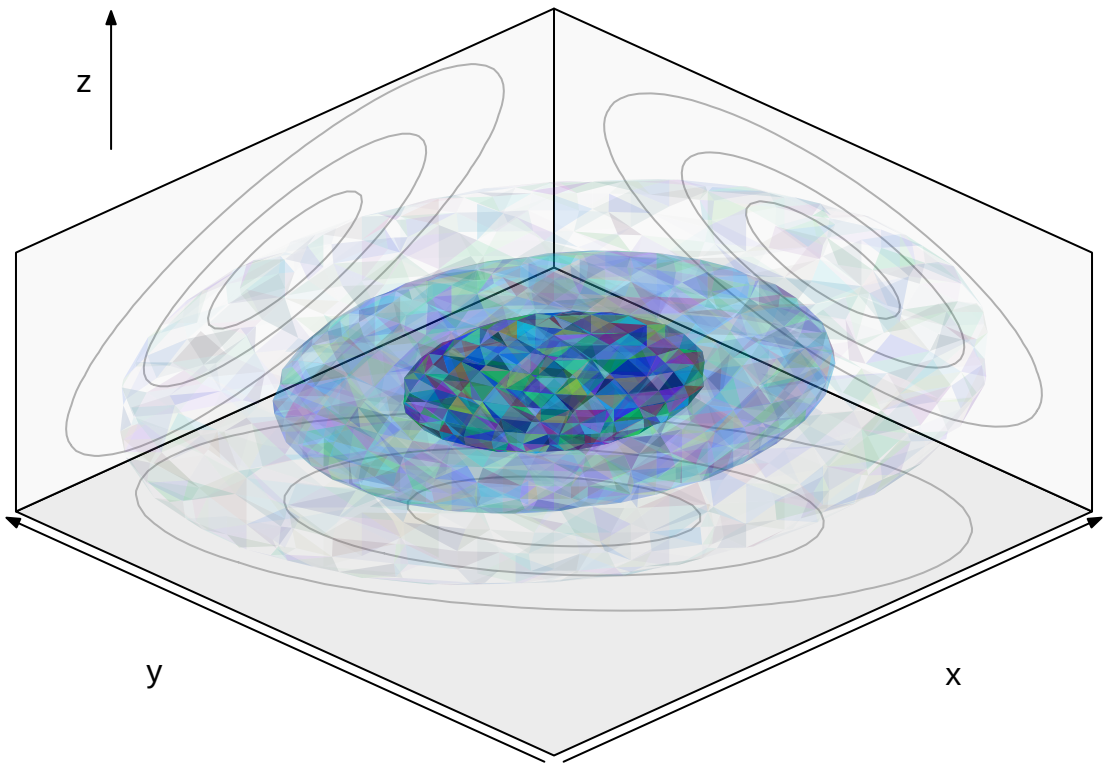
Like the bivariate normal,the mvtnorm package to evaluate trivariate normal distributions is used. We can construct a probability density function for the trivariate normal distribution using the ntvpdf or ntvpdf.2 functions as before. Both functions take nine parameters. The first function (with no suffix) take the means of X, Y and Z, the standard deviations of X, Y and Z, and their correlations cor (X; Y), cor (X;Z) and cor (Y;Z). Here's an example with different correlation coefficients:

```
f <- ntvpdf (
0, 0, 0, #mean: X, Y, Z
1, 1, 1, #sd: X, Y, Z
-0.5, 0.5, 0) #cor: X~Y, X~Z, Y~Z
f %$% covariance.matrix
```

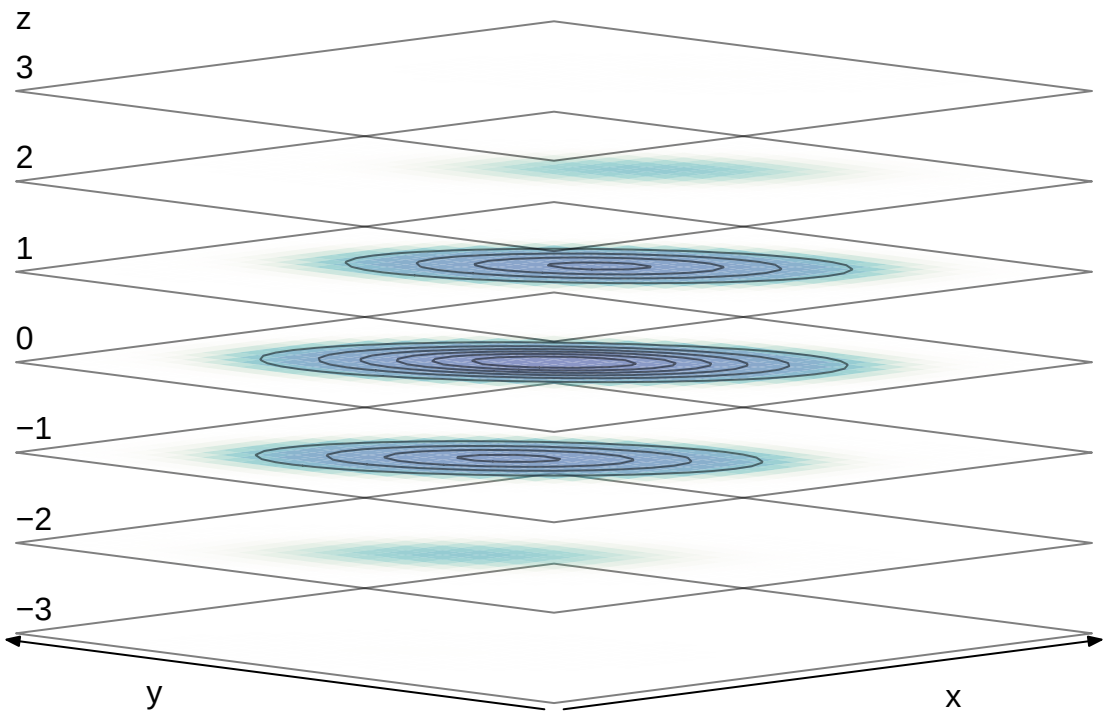
```
## Warning in f %$% covariance.matrix: %$% operator deprecated, use @
```

```
##      X    Y   Z
## X  1.0 -0.5 0.5
## Y -0.5  1.0 0.0
## Z  0.5  0.0 1.0
```

```
plot (f)
```



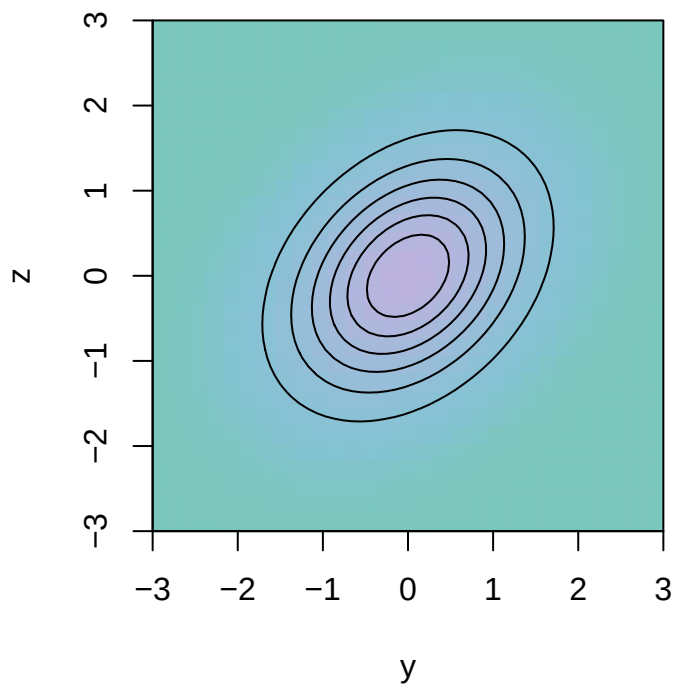

```
plot (f, FALSE)
```



Note that the panel contours represent trivariate density values, where one variable (perpanel) is held constant. They do not represent the bivariate-marginal distributions. Also, note the orientation of the plot.(Increasing y values, run diagonally from bottom-right to top-left).

The following plot, gives the trivariate density values, where x is held constant at zero:

```
f <- ntvpdf (
0, 0, 0, #mean: X, Y, Z
1, 1, 1, #sd: X, Y, Z
-0.5, 0.5, 0) #cor: X~Y, X~Z, Y~Z

plotf_cfield (function (y, z) f (0, y, z), c (-3, 3),
xlab="y", ylab="z")
```



This is equivalent the to right-rear panel from the isosurface plot. However, the horizontal axis has the reverse orientation to the y-axis from the isosurface plot. In the trivariate distribution, there's zero correlation between Y and Z. However, in the conditional distribution, there's positive correlation between Y and Z.

Comparing Normal Distributions

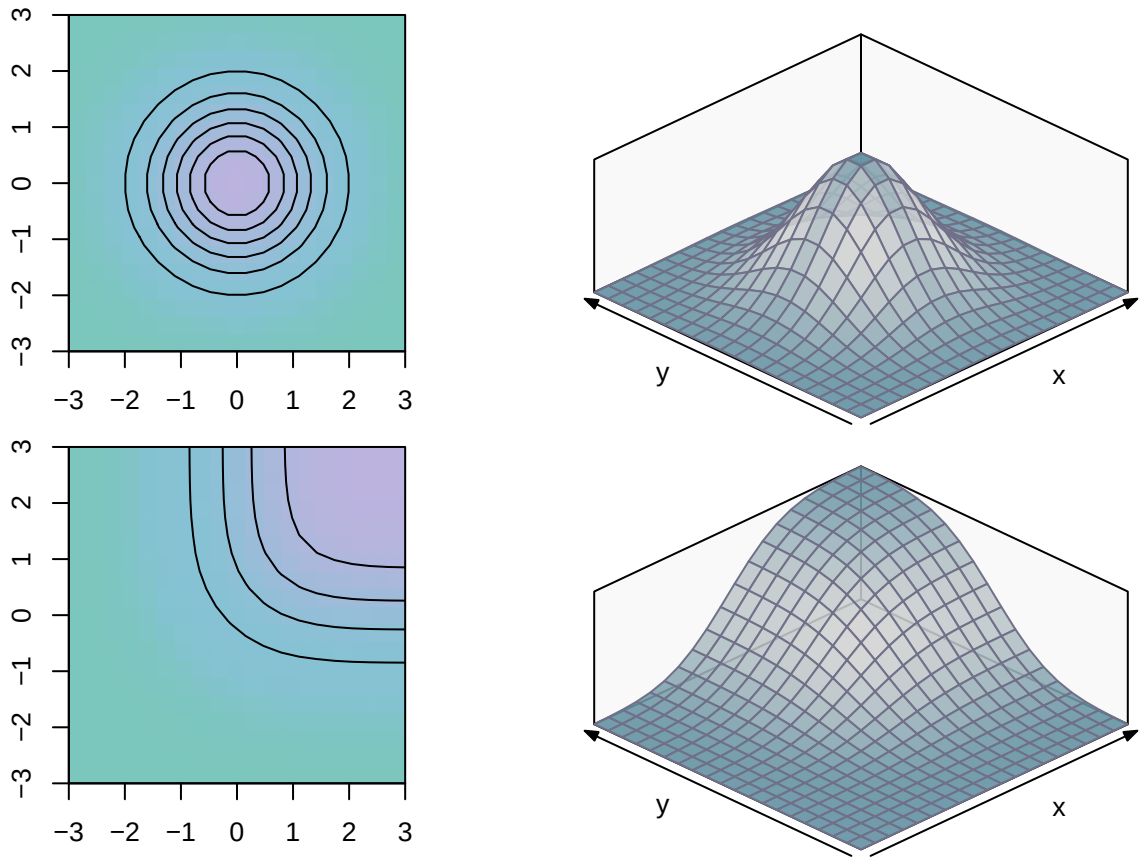
We can compare different normal distributions using different correlation or covariance parameters. First, let's consider the bivariate distribution from the earlier section with zero correlation:

```
f1 <- nbvpdf (0, 0, 1, 1, 0)
f1 %$% covariance.matrix
```

```
## Warning in f1 %$% covariance.matrix: %$% operator deprecated, use @
```

```
##   X Y
## X 1 0
## Y 0 1
```

```
#visualize
plot (f1, all=TRUE, n=20)
```



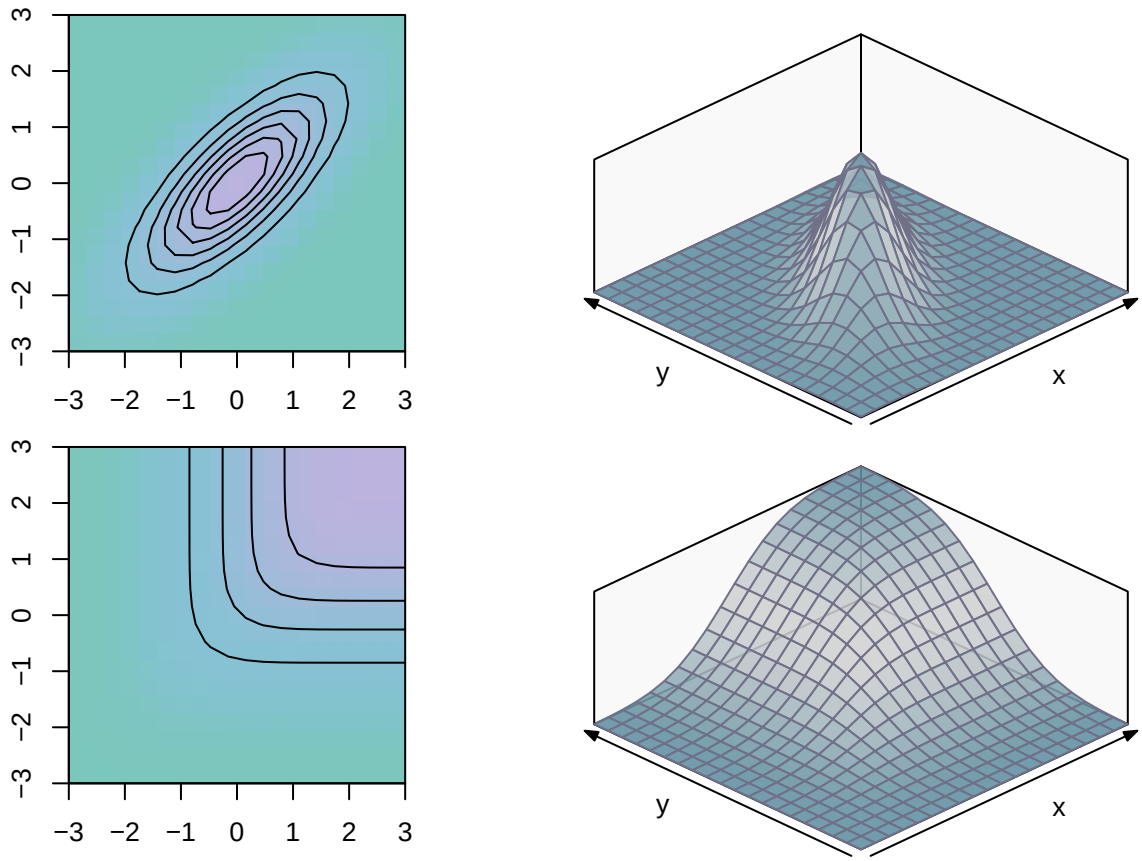
Second, let's consider bivariate distributions with positive correlation: (Note that if both the standard deviations are one, then the correlation will equal the covariance).

```
f2 <- nbvpdf (0, 0, 1, 1, 0.75)
f2 %$% covariance.matrix
```

```
## Warning in f2 %$% covariance.matrix: %$% operator deprecated, use @
```

```
##      X    Y
## X 1.00 0.75
## Y 0.75 1.00
```

```
#visualize

plot (f2, all=TRUE, n=20)
```



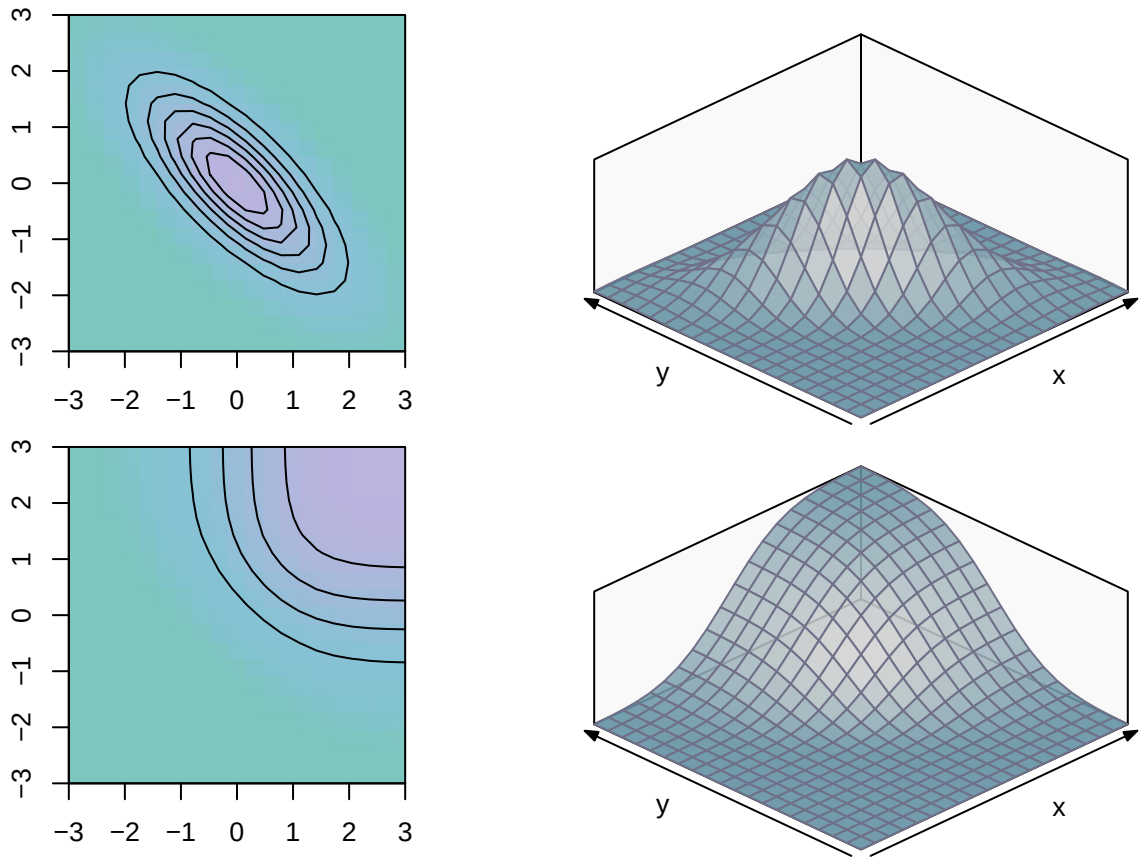
Third, let's consider bivariate distributions with negative correlation:

```
f3 <- nbvpdf (0, 0, 1, 1, -0.75)
f3 %$% covariance.matrix
```

```
## Warning in f3 %$% covariance.matrix: %$% operator deprecated, use @
```

```
##       X     Y
## X  1.00 -0.75
## Y -0.75  1.00
```

```
#visualize

plot (f3, all=TRUE, n=20)
```



Something fancy

```
#install.packages("threejs")
#install.packages("rainbow")
library(threejs)
```

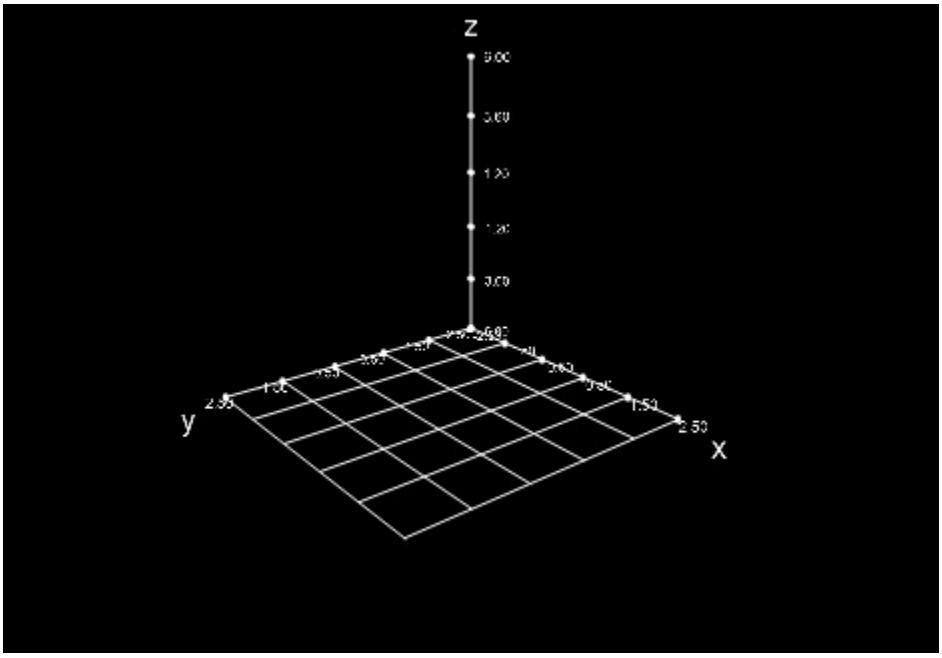
```
## Loading required package: igraph

## Warning: package 'igraph' was built under R version 4.1.2

##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##     decompose, spectrum

## The following object is masked from 'package:base':
##
##     union
```

```
library(rainbow)
```

```
## Loading required package: MASS
```

```
## Warning: package 'MASS' was built under R version 4.1.2

## Loading required package: pcaPP
```

```r
colors <- rev(rainbow(20))



N     = 20000
theta = runif(N)*2*pi
phi   = runif(N)*2*pi
R     = 1.5
r     = 1.0

x = (R + r*cos(theta))*cos(phi)
y = (R + r*cos(theta))*sin(phi)
z = r*sin(theta)

d = 6
h = 6
t = 2*runif(N) - 1
w = t^2*sqrt(1-t^2)
x1 = d*cos(theta)*sin(phi)*w
y1 = d*sin(theta)*sin(phi)*w

colorscale = cbind(seq(0, 1, by=1/(length(z) - 1)), rainbow(length(z)))

i = order(phi)
j = order(t)
col = c(rainbow(length(phi))[order(i)],
        rainbow(length(t), start=0, end=2/6)[order(j)])

M = cbind(x=c(x, x1), y=c(y, y1), z=c(z, h*t))
scatterplot3js(M, size=0.1, color=col, bg="black", pch=".")
```



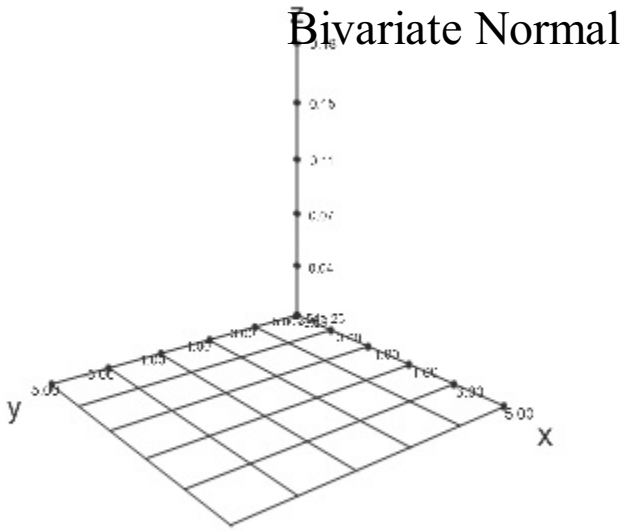

```
#something different


library(plotly)
```

```
## Warning: package 'plotly' was built under R version 4.1.2

## Loading required package: ggplot2

##
## Attaching package: 'plotly'

## The following object is masked from 'package:ggplot2':
##
##     last_plot

## The following object is masked from 'package:MASS':
##
##     select

## The following object is masked from 'package:igraph':
##
##     groups
```

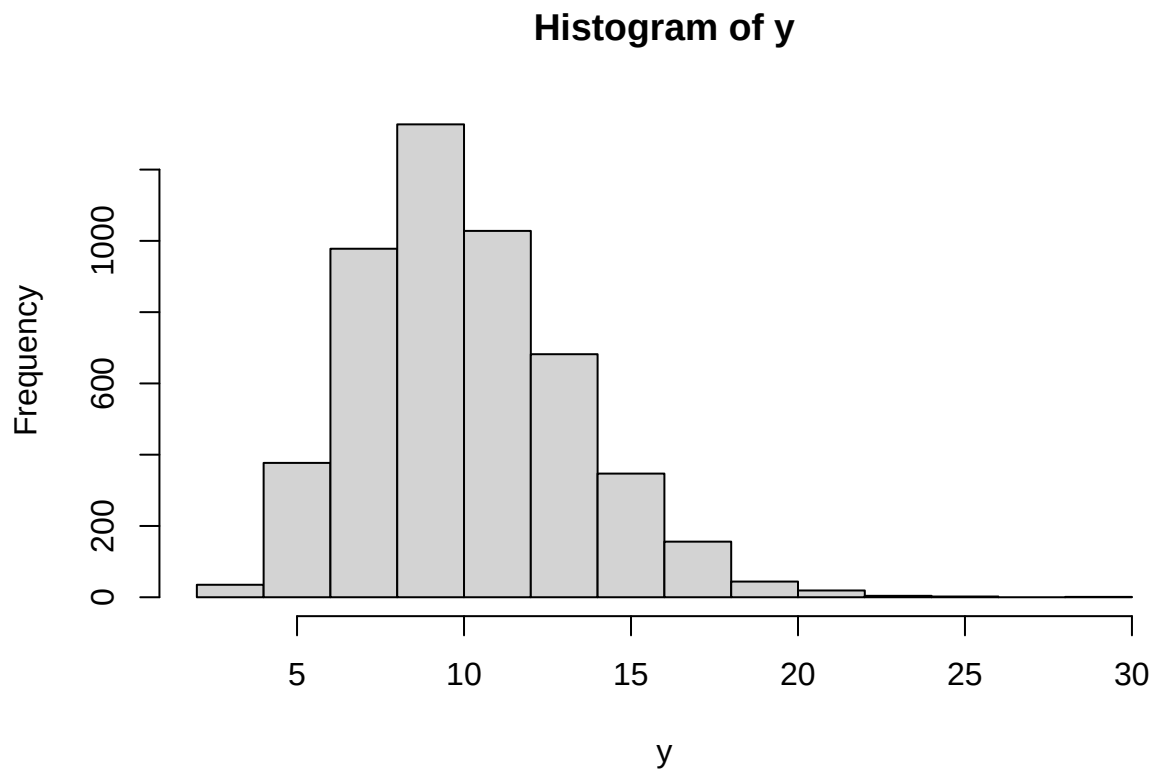
```
## The following object is masked from 'package:stats':
##
##     filter

## The following object is masked from 'package:graphics':
##
##     layout
```

```r
library(mvtnorm)
library(RColorBrewer)

u <- seq(-5, 5, by = .1)
v <- seq(-5, 5, by = .1)
M <- expand.grid(u,v)

x <- M$Var1
y <- M$Var2
sigma <- matrix(c(1, .5, .5, 1), nrow = 2, byrow = TRUE)
z <- dmvnorm(x = M, sigma = sigma)

colorscale = cbind(seq(0, 1, by=1/(length(z) - 1)), rainbow(length(z)))

scatterplot3js(x, y, z, phi = 40, theta = 20,
          color=rainbow(length(z)),
          colkey = FALSE,
          cex = .3,
          main = "Bivariate Normal")
```

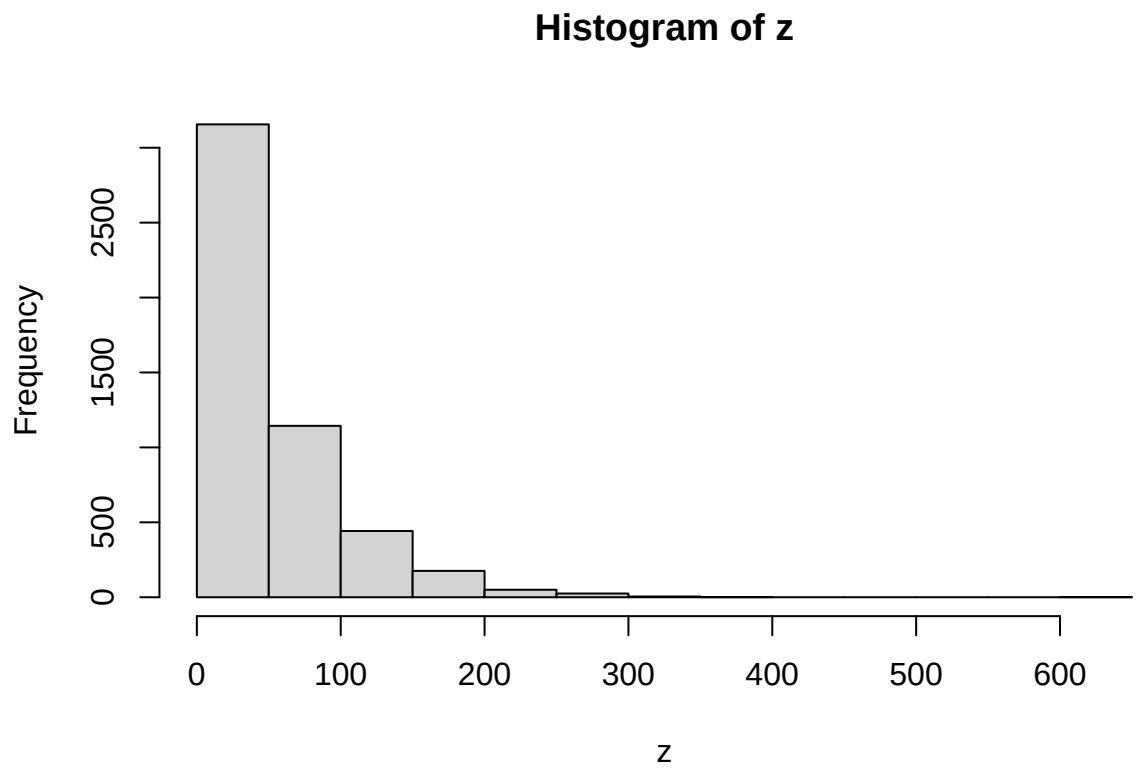
```
## Warning in ans[i] <- sprintf("%.2e", x): number of items to replace is not a
## multiple of replacement length
```




```
#Gamma distribition
set.seed(1)
y<-rgamma(n = 5000, shape = 10, scale = 1)

set.seed(1)
z<-rgamma(n = 5000, shape = 1, scale = 50)

hist(y)
```




```
hist(z)
```




```
var(z)
```

```
## [1] 2477.275
```

```
mean(z)
```

```
## [1] 50.28047
```

```
mean(y)
```

```
## [1] 10.04713
```

```
var(y)
```

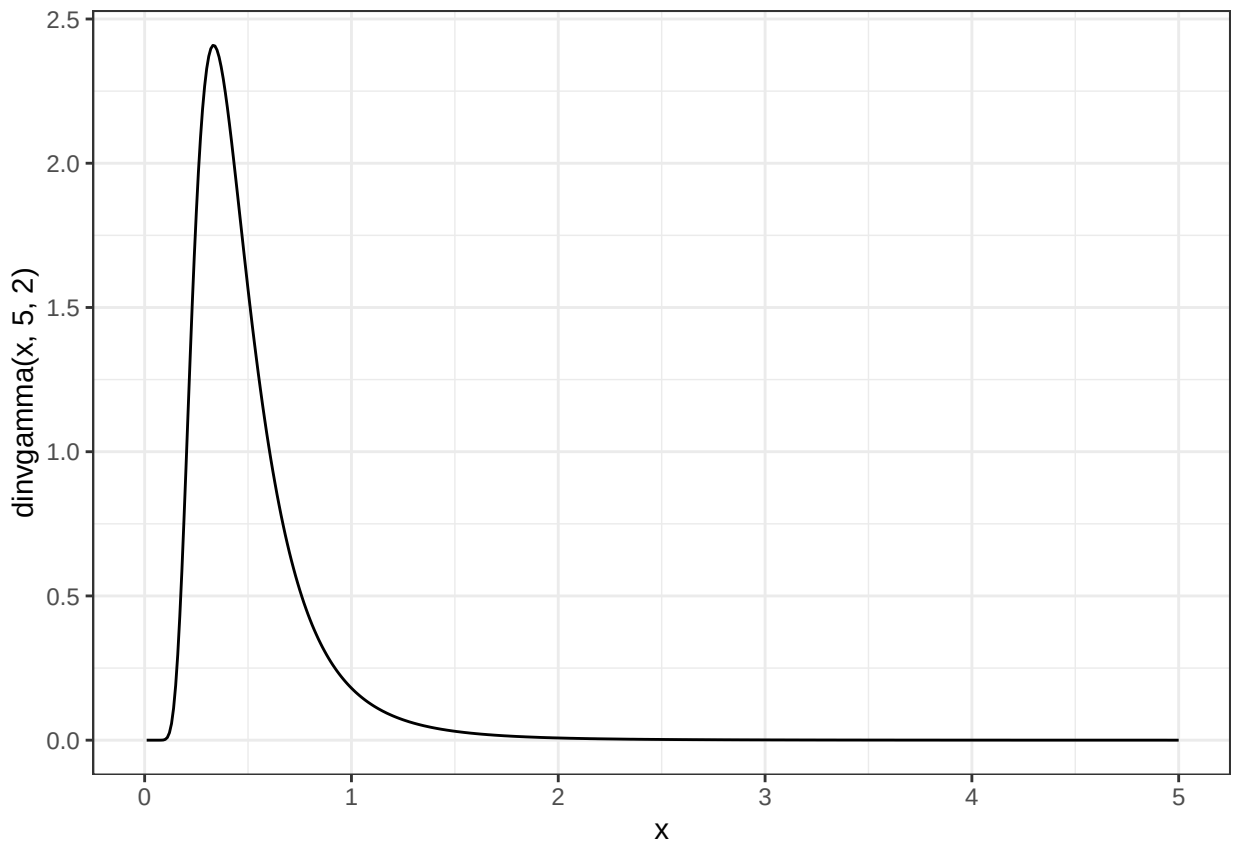
```
## [1] 10.11299
```

```
# install.packages("devtools")
devtools::install_github("dkahle/invgamma")
```

```
## Skipping install of 'invgamma' from a github remote, the SHA1 (db03bfac) has not changed since last i
##   Use `force = TRUE` to force installation
```

```r
library(invgamma)
library(ggplot2); theme_set(theme_bw())
x <- seq(0, 5, .01)
qplot(x, dinvgamma(x, 5, 2), geom = "line")
```

```
## Warning: Removed 1 row(s) containing missing values (geom_path).
```

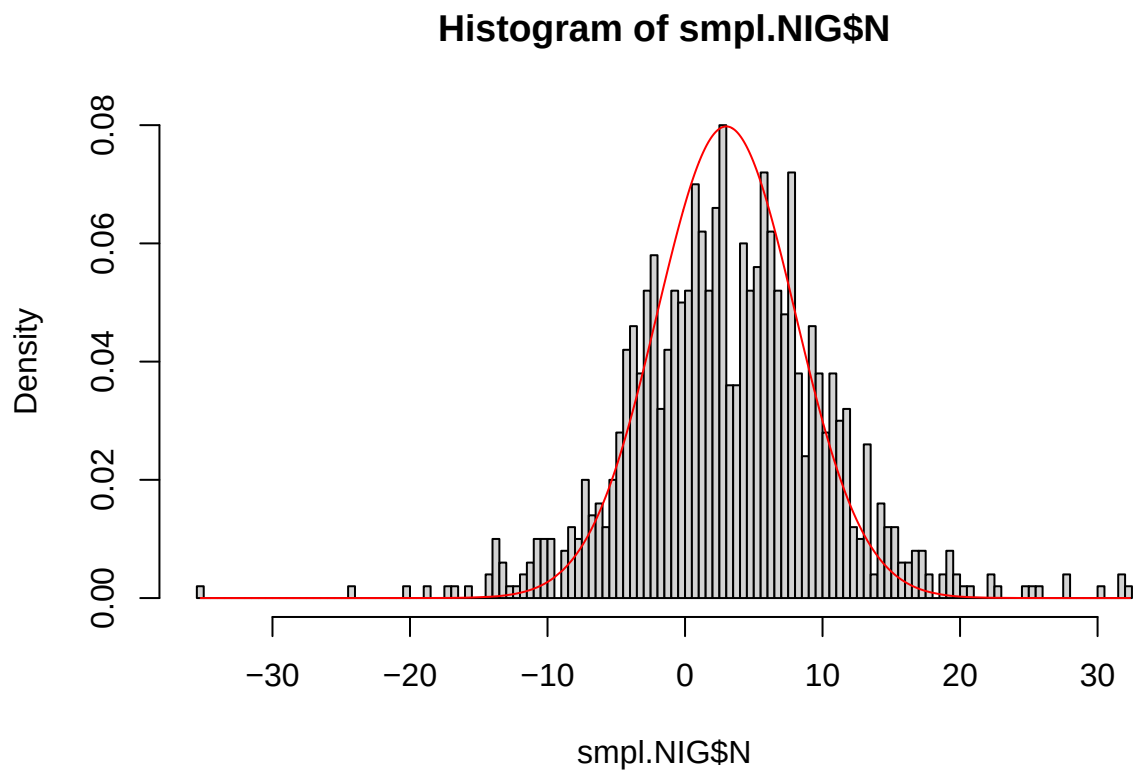



```r
#  Warning: Removed 1 rows containing missing values (geom_path).

suppressPackageStartupMessages(library(MCMCpack))
```

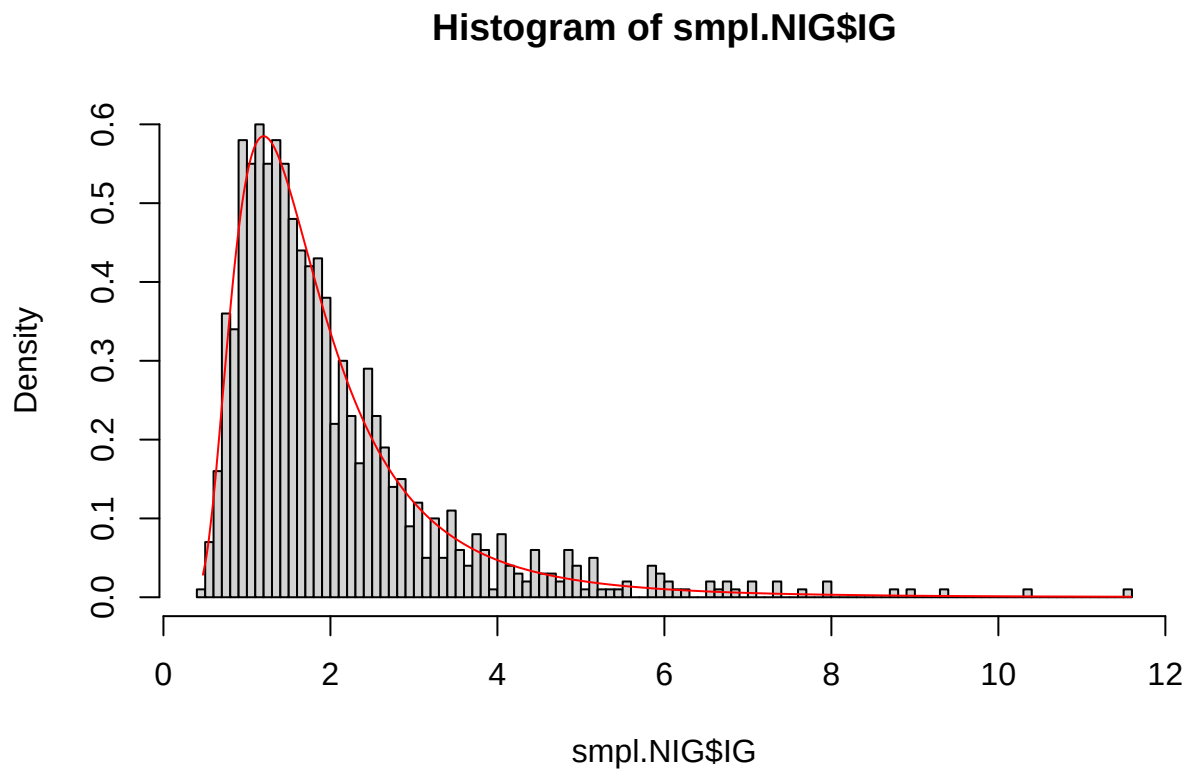
```
## Warning: package 'MCMCpack' was built under R version 4.1.2
```

```r
suppressPackageStartupMessages(library(ggplot2))

n <- 1000  #size of the sample
mu <- 3
sigma <- 5
shape <- 4
scale <- 6

#sample from inverse gamma
Y <- rinvgamma(n, shape, scale)
```

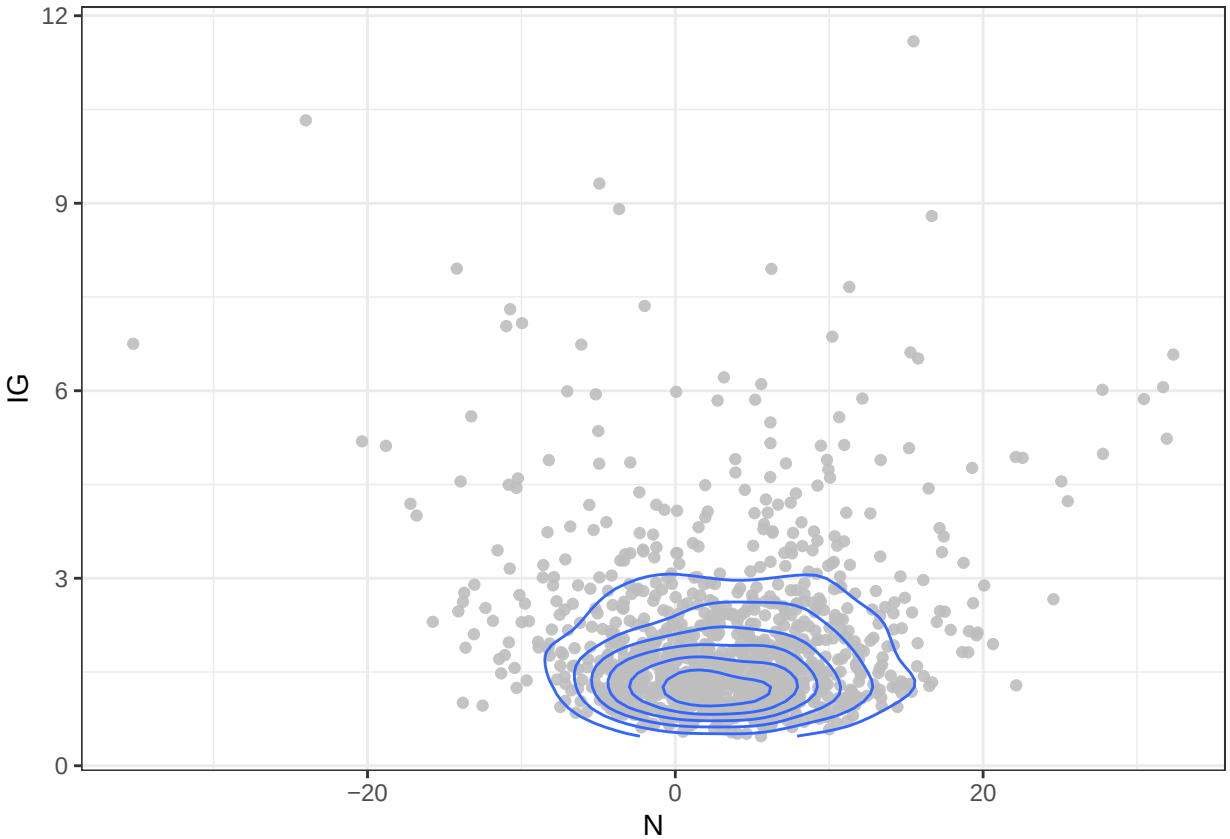
```r
#sample from X|Y distribution
X <- rnorm(n, mu, sd = sqrt(Y * sigma^2))

#Lets graph the bivariate sample we have just drawn
smpl.NIG <- data.frame(N = X, IG = Y)

#graph X
hist(smpl.NIG$N, 100, prob = TRUE)
lines(x = sort(smpl.NIG$N), y = dnorm(sort(smpl.NIG$N), mu, sigma), col = "red")
```




```r
#graph Y
hist(smpl.NIG$IG, 100, prob = TRUE)
lines(x = sort(smpl.NIG$IG), y = dinvgamma(sort(smpl.NIG$IG), shape, scale),
      col = "red")
```




```
#plot bivariate XY
p <- ggplot(data = smpl.NIG)
p <- p + geom_jitter(aes(x = N, y = IG), alpha = 0.9, colour = "grey")
p <- p + geom_density2d(aes(x = N, y = IG))
p
```



Refences

1. Spurdle, A. (2020). prohat: Multivariate Generalized Kernel Smoothing and related Statistical Methods
2. Spurdle, A. (2020). Bivariate Probability Distributions
3. Karlis, D and Ntzoufras, I. (2003). Analysis of sports data by using bivariate Poisson Models
4. Feng, D., and Tierney, L. (2013). misc3d: Miscellaneous 3D Plots